

TABLE 4.1 Differences Between Rods and Cones

	RODS	CONES
Shape	Nearly cylindrical	Tapered at one end
Prevalence in human retina	90–95%	5–10%
Greatest incidence by species	In species that are active at night	In birds, primates, and other species that are active during the day
Area of the retina	Toward the periphery	Toward the fovea
Contribution to color vision	No direct contribution	Critical for color vision
Response to dim light	Strong	Weak
Contribution to perception in detail	Little	Much

enough to find the flashlight. This *gradual improvement in the ability to see in dim light* is called **dark adaptation**.

The mechanism behind dark adaptation is this: Exposure to light causes a chemical change in certain molecules called *retinaldehydes*, thereby stimulating the visual receptors. (Retinaldehydes are derived from vitamin A.) Under normal (moderate) light, the receptors *regenerate* (rebuild) the retinaldehydes about as fast as the light is altering them, and the person maintains about a constant level of visual sensitivity. In darkness or very dim light, however, the receptors can regenerate their molecules without interruption, so the person gradually becomes better able to detect faint lights.

The cones and rods adapt to the dark at different rates. During the day our vision ordinarily relies overwhelmingly on cones. When we enter a dark place, our cones regenerate their retinaldehydes faster than the rods do, but by the time the rods finish their regeneration, they are far more sensitive to faint light than the cones are. At that point, we are seeing mostly with rods.

Here is how a psychologist demonstrates this process of dark adaptation (Goldstein, 1989): You are taken into a room that is completely dark except for one tiny flashing light. You have a knob that controls the intensity of the light; you are told to make the light so dim that you can barely see it. Over the course of three or four minutes you will gradually decrease the intensity of the light, as shown in Figure 4.8a. Note that a decrease in the intensity of the light indicates an increase in the sensitivity of your eyes. If you stare straight at the point of light, your results will demonstrate the adaptation of your cones to the dim light. (You have been focusing the light on your fovea, which has no rods.)

Now the psychologist repeats the study, with one change in procedure: You are told to stare at a very faint light while another light flashes in the periphery of your vision, where it stimulates both rods and cones. You turn a control knob until the flashing light in the periphery is just barely visible. (Figure 4.8b shows the results.) During the first 7 to 10 minutes, the results are the same as before. But then your rods become more sensitive than your cones, and you begin to see even fainter lights. Your rods continue to adapt to the dark over the next 20 minutes or so.

If you would like to demonstrate dark adaptation for yourself without any apparatus, try this: On a dark night, with only slight amounts of light coming through your windows, turn on a light in your room. Close one eye and cover it tightly with your hand for a few minutes. By the end of that time, your covered eye will be adapted to the dark and your open eye will be adapted to the light. Next, turn off your light and then open both eyes. You will see well with your dark-adapted eye and poorly with the light-adapted eye.

TRY IT YOURSELF

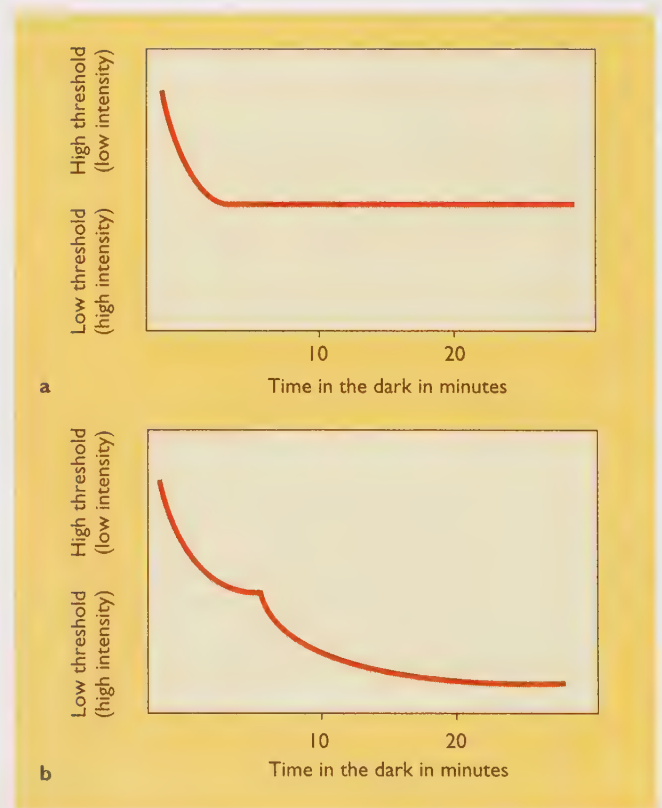


FIGURE 4.8 These graphs show dark adaptation to (a) a light you stare at directly, using only cones, and (b) a light in your peripheral vision, which you see with both cones and rods. (Based on Goldstein, 1989.)

CONCEPT CHECKS

3. You may have heard people say that cats can see in the dark. Is that possible?
4. After you have thoroughly adapted to extremely dim light, will you see more objects in your fovea or in the periphery of your eye? (Check your answers on page 117.)

The Visual Pathway

If you or I were designing an eye, we would probably run the axons of the cones and rods straight to the brain. Nature chose a different method. The visual receptors send their impulses *away from* the brain, toward the center of the eye, where they make synaptic contacts with other neurons called bipolar cells. The *bipolar cells* in turn make contact with still other neurons, the **ganglion cells**, which are *neurons that receive their input from the bipolar cells*. The *axons from the ganglion cells join to form the optic nerve, which turns around and exits the eye*, as Figures 4.2 and 4.9 show. Half of each optic nerve crosses to the opposite side of the brain at the optic chiasm (KI-az-m). Axons from the optic nerve then separate and go to several locations in the brain. In humans the largest number go to the thalamus, which then sends information to the occipital lobe, the primary area of the cortex for visual processing.

The *area where the optic nerve exits the retina* is called the **blind spot**. There is no room for receptors here because the exiting axons take up all the space. Ordinarily, you are unaware of your blind spot. To illustrate, cover your

TRY IT YOURSELF

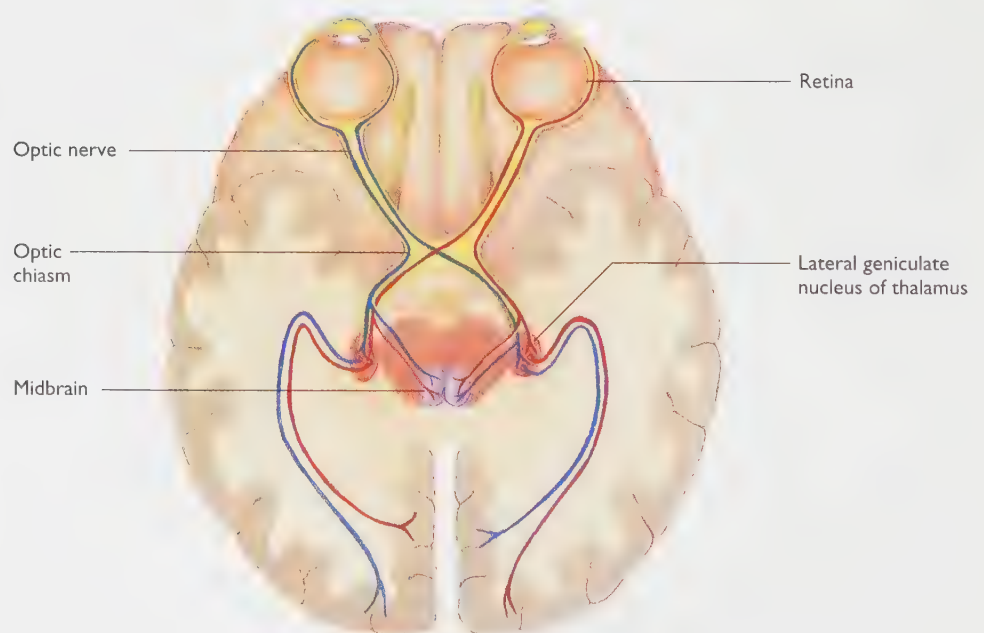
left eye and stare at the center of Figure 4.10, then slowly move the page forward and backward. When your eye is about 25–30 cm (10–12 inches) away from the page, the lion disappears, because it falls into your blind spot. What do you see in its place? Certainly not blackness. Rather, your brain makes an inference about what is in the gap. Do you *see* a continuation of the circle in that gap? Not exactly; you *infer* it more than you actually perceive it.

Information from the left and right eyes remains separate until it reaches the visual cortex. Each cell in the visual cortex receives input from one part of the left retina and a corresponding part of the right retina—that is, two retinal areas that ordinarily focus on the same point in space. Under normal conditions, the input coming from the left retina is almost the same as that coming from the right retina, and the two effects summate. However,

TRY IT YOURSELF

examine Figure 4.11 to see what happens if the retinal images conflict. Move your eyes so close to the page that the two circles seem to merge. You have some neurons in your visual cortex that respond to vertical lines. When you merge the two circles, those cells are getting stimulated by one pattern and inhibited by the other (Logothetis, Leopold, & Sheinberg, 1996). For a short while, the stimulation dominates but then for a while the inhibition dominates. Meanwhile, inhibition and stimulation are also alternating for cells that respond to horizontal lines. The net effect is that you see green lines, then red lines, then green lines again, and so forth. The *alternation between seeing the pattern in the left retina and the pattern in the right retina* is known as **binocular rivalry**.

FIGURE 4.9 Axons from cells in the retina depart the eye at the blind spot and form the optic nerve. In humans, about half the axons in the optic nerve cross to the opposite side of the brain at the optic chiasm. Some optic nerve axons carry information to the midbrain; others carry it to the thalamus, which relays information to the cerebral cortex.



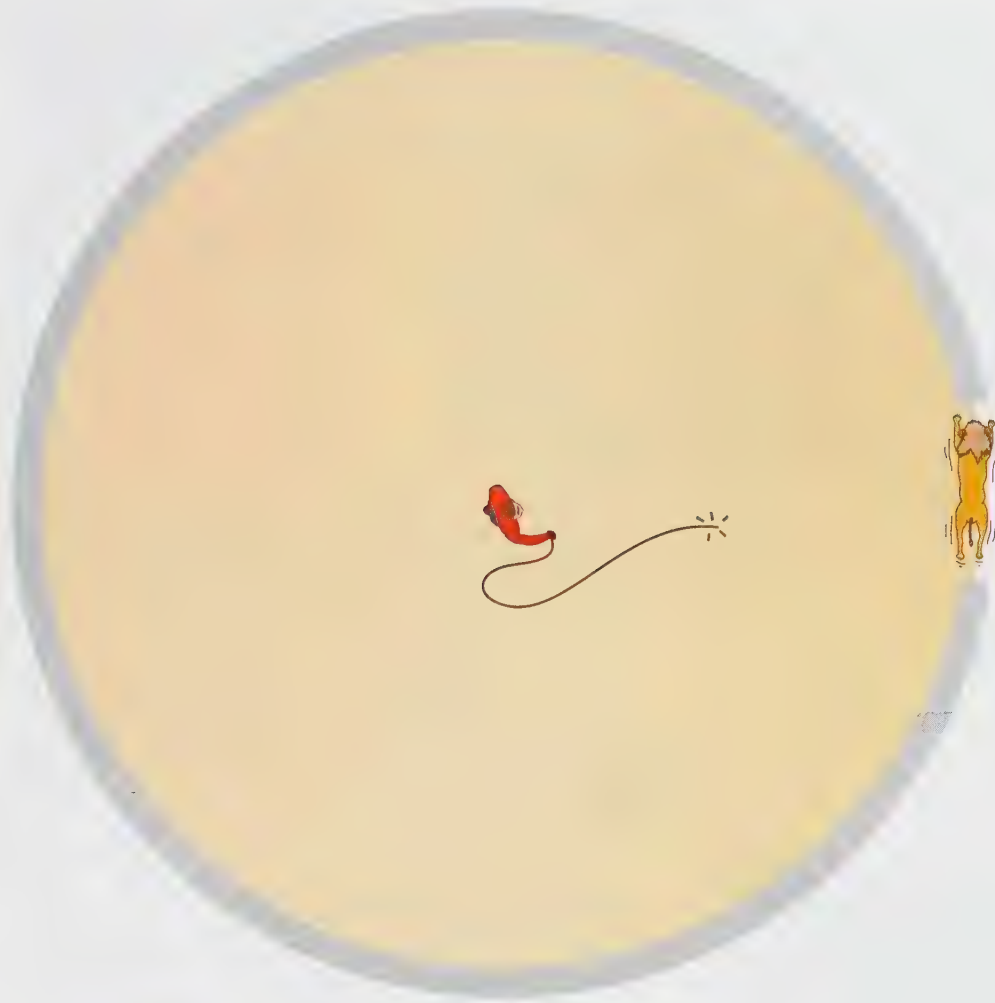


FIGURE 4.10 Close your left eye and focus your right eye on the animal trainer. Move the page toward your eyes and away from them, until you find the point where the lion on the right disappears. At that point, the lion is focused on the blind spot of your retina, where you have no receptors. What you see there is not a blank spot, but a continuation of the circle.



FIGURE 4.11 To produce binocular rivalry, move your eyes toward the page until the two circles seem to merge. You will alternate between seeing red lines and green lines.

Color Vision

As Figure 4.1 shows, different colors of light correspond to different wavelengths of electromagnetic energy. (White light consists of an equal mixture of all the visible wavelengths.) How does the visual system convert these wavelengths into

our perception of color? The process begins with three kinds of cones, which respond to different wavelengths of light. Later cells in the visual path code this wavelength information in terms of pairs of opposites—roughly, red versus green, yellow versus blue, and white versus black. Finally, cells in the cerebral cortex compare the input from various parts of the visual field to synthesize a color experience for each object. We shall examine these three stages in turn.

The Trichromatic Theory

Thomas Young was an English physician of the 1700s who, among his many other accomplishments, helped to decode the Rosetta stone (making it possible to understand Egyptian hieroglyphics), introduced the modern concept of energy, revived and popularized the wave theory of light, and offered the first theory about how people perceive color. His theory, elaborated and modified by Hermann von Helmholtz in the 1800s, came to be known as the **trichromatic theory** or the **Young-Helmholtz theory**. It is called *trichromatic* because it claims that our receptors respond to three primary colors. In modern terms, we say that *color vision depends on the*

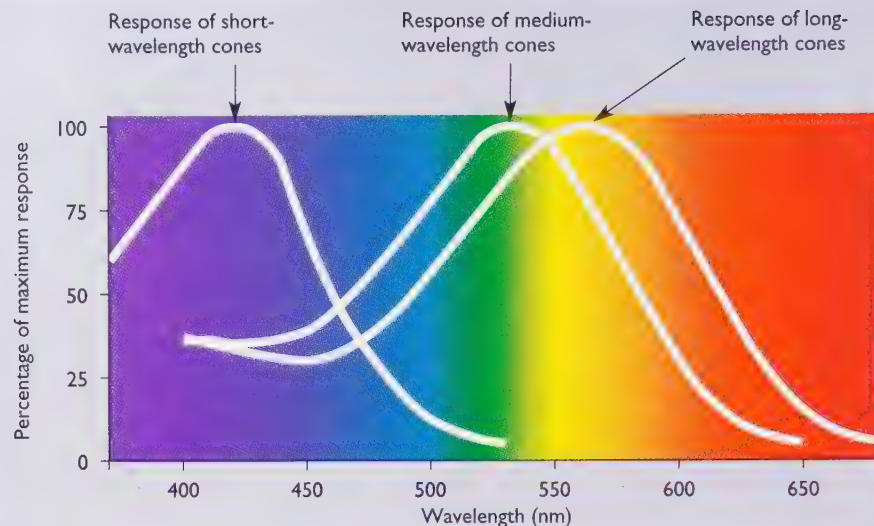


FIGURE 4.12 Sensitivity of three types of cones to different wavelengths of light. (Based on data of Bowmaker & Dartnall, 1980.)

relative rate of response by three types of cones. Each type of cone is most sensitive to a particular range of light wavelengths (Figure 4.12). One type is most sensitive to short wavelengths (which we generally see as blue), another to medium wavelengths (seen as green), and another to long wavelengths (red). Each wavelength prompts varying levels of activity in the three types of cones. So, for example, green light excites mostly the medium-wavelength cones, red light excites mostly the long-wavelength cones, and yellow light excites the medium-wavelength and long-wavelength cones about equally. Every wavelength of light produces its own distinct ratio of responses by the three kinds of cones. White light excites all three kinds of cones equally.

Young and Helmholtz proposed their theory long before experiments confirmed the existence of these three types of cones (Wald, 1968). They relied entirely on a behavioral observation: Observers can take three different colors of light and then, by mixing them in various proportions, match all other colors of light. (Note that mixing light of different colors is not the same as mixing paints of different colors. Mixing yellow and blue *paints* produces green; mixing yellow and blue *lights* produces white.)

The short-wavelength cones, which respond most strongly to blue, are less numerous than the other two types of cones, especially in the fovea. Consequently, a tiny blue

point may look black. For the retina to detect blueness, the blue must extend over a moderately large area. Figure 4.13 illustrates this effect. Count the red spots and then the blue spots. Then stand farther away and count the spots again. You will probably see as many red spots as before but fewer blue spots.

TRY IT YOURSELF

CONCEPT CHECK

- According to the trichromatic theory, how does our nervous system tell the difference between bright yellow-green and dim yellow-green light? (Check your answer on page 117.)

The Opponent-Process Theory

Young and Helmholtz were right about how many cones we have, but our perception of color has some complicated features that the trichromatic theory cannot easily handle. For example, four colors, not three, *seem* to most people to be primary or basic: red, green, yellow, and blue. Yellow simply does not seem like a mixture of reddish and greenish experiences, nor is green a yellowish blue. Furthermore, if you stare at an object of one color, say red, and then look away, you see a colored afterimage—in this case green. Each color appears to have an opposite, and there is no obvious reason why it should, according to the trichromatic theory.

For these reasons, another 19th-century scientist, Ewald Hering, proposed the **opponent-process theory** of color vision, which accounts for much of color information processing after the cones. According to this theory, *we perceive color not in terms of independent colors but in terms of a system of paired opposites: red versus green, yellow versus blue, and white versus black.* Any light stimulus leads to a perception somewhere along each of these three dimensions. Although Hering didn't know it, the dimen-

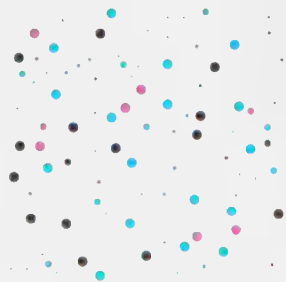


FIGURE 4.13 Blue spots look black unless they cover a sizable area. Count the red dots, then count the blue dots. Try again while standing farther away from the page.

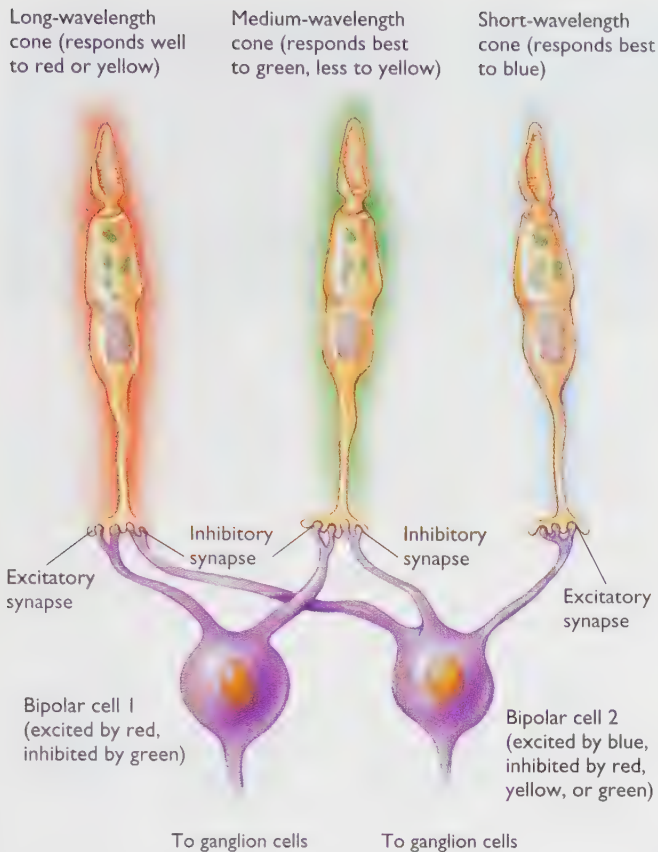


FIGURE 4.14 According to the opponent-process theory of color vision, the responses of three kinds of cones excite and inhibit bipolar cells, which relay their responses to other cells in the visual system. For example, red light excites the long-wavelength cone and thereby excites bipolar cell 1; green light excites the medium-wavelength cone and thereby inhibits bipolar cell 1. Therefore, bipolar cell 1 increases its response in the presence of red light and decreases its response in the presence of green light. After prolonged exposure to red light, the cell will “rebound” to an inhibition of response and therefore report a message of “green.”

sions correspond to excitation and inhibition of many cells in the visual system (DeValois, 1965; Michael, 1978). For example, some bipolar cells in the eye are excited when green light strikes the cones that connect to them and inhibited when red light strikes.¹ Other bipolar cells are excited by red and inhibited by green. Still other cells are excited by yellow and inhibited by blue or excited by blue and inhibited by yellow. The white/black system is more complicated: If a cell is excited by white light, we cannot meaningfully say that it is inhibited by black. Rather, the cell is inhibited when the light on neighboring areas of the retina is brighter than the light in its own area.

¹Technically, it is wrong to talk about red, green, or blue light. I should say “long-wavelength,” “medium-wavelength,” or “short-wavelength” light. The light itself is not red, green, or any other color; the color perception is in us. Furthermore, light that is predominantly medium wavelength can, under certain circumstances, be seen as a color other than green. However, to simplify and speed up the conversation, I use terms like *red light* to refer to light that we usually see as red.

For example, in Figure 4.14, bipolar cell 1 receives an excitatory synaptic message from the long-wavelength cone and an inhibitory synaptic message from the medium-wavelength cone. The cell increases its response in the presence of red light and decreases its response in the presence of green light. Bipolar cell 2 receives excitatory synaptic messages from the short-wavelength cone; it is therefore excited by blue light. Cell 2 receives inhibitory messages from both the long-wavelength cone and the medium-wavelength cone; it will be inhibited by either red, yellow, or green light.

TRY IT YOURSELF

Figure 4.15 lends support to the opponent-process theory. Stare at the white dot near the center of the figure and stare at it for a minute or so, preferably under a bright light, without moving your eyes or your head. Then look at a plain white or gray background. *Do this now.*

If you have normal or near-normal vision, you saw the Mona Lisa when you looked away. After bipolar cells and other cells in your visual system have been stimulated long enough they become fatigued, and removal of the stimulus will leave them inhibited. Recall that a cell that is excited by red light is inhibited by green light, and vice versa. Thus, after the removal of a prolonged red light, the cells are responding just as they do in the presence of green light, so



FIGURE 4.15 Use this image to see the negative afterimages of opposite colors, which rebound after sufficient stimulation. Stare at the white dot for a minute or more, then focus on a white background.



FIGURE 4.16 Despite the green and red filters used in producing the center and right-hand photographs, you can still identify the colors of the objects in the photos. The effect would be much more convincing if you were actually on location, looking at this scene through tinted glasses. Here, each photo is surrounded by the white page, which sets an effective standard. Your ability to identify the color of an object, despite changes in the light striking the environment, is called **color constancy**.

you in fact see green. Similarly, after the removal of a green light you see red, after yellow you see blue, and after blue you see yellow. These *experiences of one color after the removal of another* are called **negative afterimages**.

CONCEPT CHECKS

6. How would bipolar cell I in Figure 4.14 respond to yellow light? Why?
7. The negative afterimage that you created by staring at Figure 4.15 may seem to move against the background. Why doesn't it stay in one place? (Check your answers on page 117.)

The Retinex Theory

The opponent-process theory, though it accounts for many phenomena of color vision, overlooks one important one. Suppose you look at a large white screen illuminated entirely with green light, in an otherwise dark room. How would you know whether this is a white screen illuminated with green light or a green screen illuminated with white light? or a blue screen illuminated with yellow light? (Actually, the possibilities go on and on.) The answer is, you would not know. But now someone wearing a brown shirt and blue jeans stands in front of the screen. By comparing what you see of this person to what you see on the screen, you immediately see the shirt as brown, the jeans as blue, and the screen as white, even though all these objects are reflecting more green light than anything else. The point is that we do not ordinarily perceive the color of an object in isolation. We perceive the color of any object by comparing the light it reflects to the light that other objects in that scene reflect. As a result, we can perceive blue jeans as blue and bananas as yellow regardless of whether we are in dim light or bright light, outdoors or indoors, or wearing tinted glasses. This *tendency of an object to appear nearly the same color under a variety of lighting conditions* is called **color constancy**. (See Figure 4.16.)

In response to such observations, Edwin Land (the inventor of the Polaroid Land Camera) proposed the **retinex**

theory. According to this theory, *we perceive color through the cerebral cortex's comparison of various retinal patterns* (Figure 4.17). (*Retinex* is a combination of the words *retina* and *cortex*.) The cerebral cortex compares the patterns of light coming from different areas of the retina and synthesizes a color perception for each area (Land, Hubel, Livingstone, Perry, & Burns, 1983; Land & McCann, 1971).

The strongest evidence for this theory comes from brain-damaged patients. Damage to one region of the occipital cortex destroys color constancy (Wild, Butler, Carden, & Kulikowski, 1985; Zeki, 1993). To a person with such brain damage, an object that looks orange under one light looks red under another; yellow, greenish, or even white under still other lights. As the retinex theory predicts, the phenomenon of color constancy depends on the activity of the cerebral cortex.

In the 1800s, the trichromatic theory and the opponent-process theory were considered rival theories, but vision researchers today consider both of them, as well as the retinex theory, to be correct statements that happen to address different stages of visual processing. The trichromatic theory is certainly correct in stating that human color vision starts with three kinds of cones. The opponent-process theory explains how the bipolar cells and later cells organize color information. The retinex theory adds the final touch, noting that the cerebral cortex compares color information from various parts of the visual field.

SOMETHING TO THINK ABOUT

If you stare for a minute at a small green object on a white background and then look away, you will see a red afterimage. But if you stare at a green wall nearby so that you see nothing but green in all directions, then when you look away you do not see a red afterimage. Why not? ★

Colorblindness

For a long time, people apparently assumed that anyone with normal vision could see and recognize colors

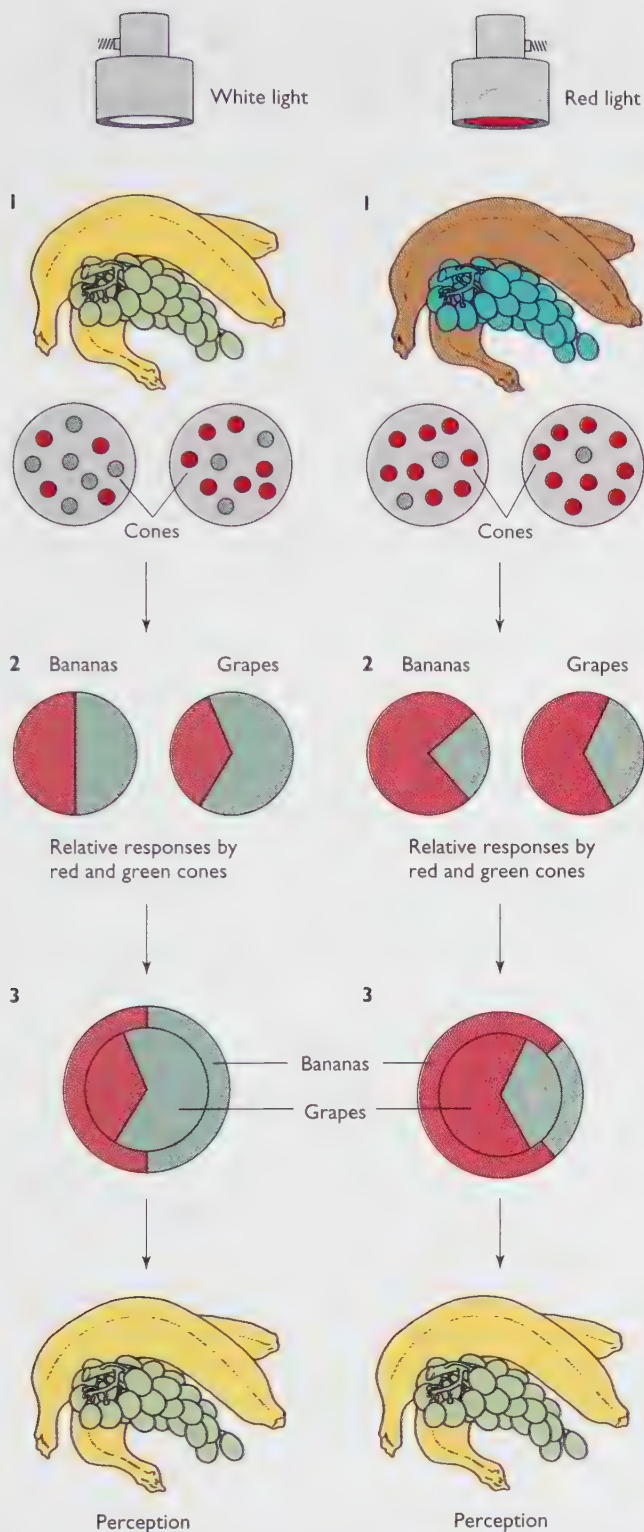


FIGURE 4.17 When bananas and grapes reflect red light, they excite a higher percentage of long-wavelength (red) cones than usual. According to the retinex theory, brain cells determine the red-green percentage for each fruit. Then cells in the visual cortex divide the “red-greenness” of the bananas by the “red-greenness” of the grapes to produce color sensations. In red and white light, the ratios between the fruits are nearly constant.

(Fletcher & Voke, 1985). Then, during the 1600s, the phenomenon of colorblindness (or color deficiency) was un-

ambiguously recognized. Here was the first clue that color vision is a function of our eyes and brains, and not just of the light itself.

The total inability to distinguish one color from another is extremely rare, except as a result of brain damage. However, about 4% of all people are partially colorblind. Investigators believe that most cases of colorblindness result from either the absence of one of the three types of cones or a decreased responsiveness by one of those types (Fletcher & Voke, 1985). People with a deficiency of the medium-wavelength cones are relatively insensitive to medium-wavelength (green) light. I do not mean that such people do not see such light; they perceive a green patch as almost gray, especially if the patch is small (Boynton, 1988).

The most common type of colorblindness is sometimes known as **red-green colorblindness**. People with red-green colorblindness *have difficulty distinguishing red from green and either red or green from yellow*. Actually, red-green colorblindness has two forms, *protanopia* and *deuteranopia*. People with protanopia lack long-wavelength cones; people with deuteranopia lack medium-wavelength cones. People with the rare *yellow-blue colorblindness* (also known as *tritanopia*) have trouble distinguishing yellows and blues. They are believed to lack short-wavelength cones.

TRY IT YOURSELF

Figure 4.18 gives a crude but usually satisfactory test for red-green colorblindness. What do you see in each part of the figure? (To interpret your answers, refer to answer A on page 118.)

How does the world look to colorblind people? Their descriptions use all the usual color words: Roses are red, violets are blue, bananas are yellow, grass is green. But this does not mean that they perceive colors the same as does a person with normal color vision. Can they tell us what that “red” rose actually looks like to them? In most cases, no. Certain rare individuals, however, are red-green colorblind in one eye but have normal vision in the other eye. Because they know what the color words really mean (from experience with their normal eye), they can tell us what their colorblind eye sees. They say that objects that look red or green to the normal eye look yellow or yellow-gray to the colorblind eye (Marriott, 1976).

If you have normal color vision, Figure 4.19 will show you what it is like to be red-green colorblind. First cover part b, a typical item from a colorblindness test, and stare at part a, a red field, under a bright light for about a minute. (The brighter the light and the longer you stare, the greater the effect will be.) Then look at part b. Staring at the red field has fatigued your red cones, so you will now have only a weak sensation of red. As the red cones recover, you will see part b normally.

TRY IT YOURSELF

Now stare at part c, a green field, for about a minute and look at part b again. Because you have fatigued your green cones, the figure in b will stand out even more strongly than usual. In fact, certain red-green colorblind people may be able to see the number in b only after staring at c. (Refer to answer B on page 118.)

THE MESSAGE

How We See

Before there were any people or other color-sighted animals, was there any color on Earth? *No*. There was light, to be sure, and different objects reflected different wavelengths of light, but color exists only in brains, not in the objects themselves. Our vision is not just a copy of the outside world; it is a construction that enables us to interact with the world to our benefit.

To many readers, vision and our other senses seem more complicated than some of the later topics in this book, such as emotion. To some extent sensation may really be more complicated than emotion, but the apparent difference in complexity exists partly because researchers have learned so much more about the senses. The more we learn about any topic, the more sophisticated questions we can ask. The complexity of what we now know about vision is a tribute to many generations of researchers.

SUMMARY

- ✳ **Common misconceptions.** The eyes do not send out “sight rays,” nor does the brain build little copies of the stimuli it senses. It converts or translates sensory stimuli into an arbitrary code that represents the information. (page 105)
- ✳ **Focus.** The cornea and lens focus the light that enters through the pupil of the eye. If the eye is not spherical or if the lens is not flexible, corrective lenses may be needed. (page 106)
- ✳ **Cones and rods.** The retina contains two kinds of receptors: cones and rods. Cones are specialized for detailed vision and color perception. Rods detect dim light. (page 107)
- ✳ **Blind spot.** The blind spot is the area of the retina through which the optic nerve exits; this area has no visual receptors and is therefore blind. (page 110)
- ✳ **Binocular rivalry.** Under normal circumstances, the two retinas receive similar information patterns and provide the brain with information for three-dimensional perception. If the two retinas receive incompatible patterns, we experience a competition in which one pattern or the other dominates at any given moment. (page 110)



FIGURE 4.18 These items provide an informal test for red-green colorblindness, an inherited condition that mostly affects men. What do you see? Compare your answers to answer A, page 118.

SOMETHING TO THINK ABOUT

In the introduction to this chapter I suggested that we would see no new colors on another planet and that we cannot be certain that different people on Earth really have the same color experiences. Now try to explain the reasons behind those statements. ✳

FIGURE 4.19 These stimuli induce temporary red-green colorblindness and temporarily enhance color vision. First stare at pattern a under a bright light for about a minute, then look at b. What do you see? Next stare at c for a minute and look at b again. Now what do you see? Compare your answer to answer B, page 118.



- ★ **Color vision.** Color vision depends on three types of cones, each most sensitive to a particular range of light wavelengths. The cones transmit messages so that the bipolar and ganglion cells in the visual system are excited by light of one color and inhibited by light of the opposite color. Then the cerebral cortex compares the responses from different parts of the retina to determine the color of light coming from each area of the visual field. (page 111)
- ★ **Colorblindness.** Complete colorblindness is rare. Certain people have difficulty distinguishing reds from greens; in rare cases, some have difficulty distinguishing yellows from blues. (page 114)

Suggestion for Further Reading

Hubel, D. H. (1988). *Eye, brain, and vision*. New York: Scientific American Library. A treatment by an investigator who shared the Nobel Prize in physiology and medicine for his research on the physiology of vision.

Terms

- sensation** the conversion of energy from the environment into a pattern of response by the nervous system (page 104)
- perception** the interpretation of sensory information (page 104)
- stimulus** energy in the environment that affects what we do (page 105)
- receptor** a specialized cell that converts environmental energies into signals for the nervous system (page 105)
- electromagnetic spectrum** the continuum of all the frequencies of radiated energy (page 105)
- pupil** the adjustable opening in the eye through which light enters (page 105)
- iris** the colored structure on the surface of the eye, surrounding the pupil (page 105)
- retina** a layer of visual receptors covering the back surface of the eyeball (page 106)
- cornea** a rigid, transparent structure on the surface of the eyeball (page 106)
- lens** a flexible structure that can vary its thickness to enable the eye to focus on objects at different distances (page 106)
- accommodation of the lens** adjustment of the thickness of the lens to focus on objects at different distances (page 106)
- presbyopia** decreased flexibility of the lens and therefore inability to focus on nearby objects (page 106)
- myopia** nearsightedness; the inability to focus on distant objects (page 106)
- hyperopia** farsightedness; the inability to focus on nearby objects (page 106)
- glaucoma** a condition characterized by increased pressure within the eyeball, resulting in damage to the optic nerve and therefore a loss of vision (page 106)
- cataract** a disorder in which the lens of the eye becomes cloudy (page 106)
- cone** the type of visual receptor that is adapted for color vision, daytime vision, and detailed vision (page 107)
- rod** the type of visual receptor that is adapted for vision in dim light (page 107)

- fovea** the central part of the retina that has a greater density of receptors, especially cones, than any other part of the retina (page 107)
- dark adaptation** a gradual improvement in the ability to see in dim light (page 109)
- ganglion cells** neurons in the eye that receive input from bipolar cells, which in turn receive their input from the visual receptors (page 110)
- optic nerve** a set of axons that extend from the ganglion cells of the eye to the thalamus and several other areas of the brain (page 110)
- blind spot** the area where the optic nerve exits the retina (page 110)
- binocular rivalry** alteration between seeing the pattern in the left retina and the pattern in the right retina (page 110)
- trichromatic theory** or **Young-Helmholtz theory** the theory that color vision depends on the relative rate of response of three types of cones (page 111)
- opponent-process theory** (of color vision) the theory that we perceive color in terms of a system of paired opposites: red versus green, yellow versus blue, and white versus black (page 112)
- negative afterimage** a color that a person sees after staring at its opposite color for a while (page 114)
- color constancy** the tendency of an object to appear nearly the same color under a variety of lighting conditions (page 114)
- retinex theory** the theory that color perception results from the cerebral cortex's comparison of various retinal patterns (page 114)
- red-green colorblindness** impaired ability to distinguish red from green and either red or green from yellow (page 115)

Answers to Concept Checks

1. If your vision is normal, then wearing glasses intended for a myopic person will make your vision blurry. Such glasses alter the light as though they were bringing the object closer to the viewer. Unless the glasses are very strong, you may not notice much difference when you are looking at distant objects, because you can adjust the lens of your eyes to compensate for what the glasses do. However, nearby objects will appear blurry in spite of the best compensations that the lenses of your eyes can make. (page 107)
2. The center of the retina consists entirely of cones. If you look slightly to the side, the light falls on an area of the retina that consists partly of rods, which are more sensitive to faint light. (page 108)
3. As do people, cats can adapt well to dim light. No animal, however, can see in complete darkness. Vision is the detection of light that strikes the eye. (Similarly, the X-ray vision attributed to the comic book character Superman is impossible. Even if he could send out X rays, he would not see anything unless those X rays bounced off an object and back into his eyes. (page 110)
4. You will see more objects in the periphery of your eye. The fovea contains only cones, which cannot become as sensitive as the rods do in the periphery. (page 110)
5. Although bright yellow-green and dim yellow-green light would evoke the same ratio of activity by the three cone

types, the total amount of activity would be greater for the bright yellow-green light. (page 112)

6. Ganglion cell 1 would be almost unaffected by yellow light. Yellow light would stimulate the long-wavelength cone, which excites ganglion cell 1, but it would stimulate the medium-wavelength cone, which inhibits ganglion cell 1, about equally. (page 113)
7. The afterimage is on your eye, not on the background. When you try to focus on a different part of the afterimage, you move your eyes and the afterimage moves with them. (page 114)

Answers to Other Questions in the Text

- A. In Figure 4.18a, a person with normal color vision sees the numeral 74; in Figure 4.18b, the numeral 8.
- B. In Figure 4.19b, you should see the numeral 29. After you have stared at the red circle in part a, the 29 in b may look less distinct than usual, as though you were red-green color-

blind. After staring at the green circle, the 29 may be even *more* distinct than usual. If you do not see either of these effects at once, try again, but this time stare at a or c a little longer, *and* continue staring at b a little longer. The effect does not appear immediately, only after a few seconds.

Web Resources

IllusionWorks

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The Nonvisual Senses

How do hearing, the vestibular sense, skin senses, pain, taste, and olfaction work?

Consider these common expressions:

I *see* what you mean.

I *feel* sympathy toward your plight.

I am deeply *touched* by everyone's support and concern.

The Senate will *hold* hearings on the budget proposal.

She is a person of great *taste*.

He was *dizzy* with success.

The policies of this company *stink*.

That *sounds* like a good job offer.

Each sentence expresses an idea in terms of sensation, though we know that these terms are not meant to be taken literally. If you compliment people on their “fine taste,” you are not referring to their tongues.

The broad, metaphorical use of terms of sensation is not accidental. Our thinking and brain activity deal mostly, if not entirely, with sensory stimuli. Perhaps you doubt that assertion: “What about abstract concepts?” you might object. “Sometimes I think about numbers, time, love, justice, and all sorts of other nonsensory concepts.” Yes, but how did you learn those concepts? Didn't you learn numbers by counting objects you could see or touch? Didn't you learn about time by observing changes in sensory stimuli? Didn't you learn about love and justice from specific events that you saw, heard, and felt? Could you explain any abstract concept without referring to something you detect through your senses?

We have already considered how we detect light. Now let's discuss how we detect sounds, head tilt, skin stimulation, and chemicals.

Hearing

Fish detect vibrations in the water by means of a long row of touch receptors along their sides, called the *lateral line system*. The mammalian ear, which probably evolved as a

modification of the lateral line system, converts sound waves into mechanical displacements of a membrane that a row of receptor cells can detect.

Sound waves are *vibrations of the air or of another medium*. They vary in both frequency and amplitude (Figure 4.20). The frequency of a sound wave is the number of *cycles (vibrations) that it goes through per second*, designated **hertz (Hz)**. **Pitch** is a *perception closely related to frequency*. We perceive a high-frequency sound wave as high pitched and a low-frequency sound as low pitched. **Loudness** is a *perception that depends on the amplitude of a sound wave*—the vertical range of its cycles. Other things being equal, the greater the *amplitude* of a sound, the louder it sounds to us. Because pitch and loudness are psychological experiences, however, they are influenced by factors other than the physical frequency and amplitude of sound waves. For example, tones of different frequencies may not sound equally loud, even though they have the same physical amplitude.

The ear, a complicated organ, converts relatively weak sound waves into more intense waves of pressure in the *fluid-filled canals of the snail-shaped organ* called the **cochlea** (KOCK-lee-uh), *which contains the receptors for hearing* (Figure 4.21). When sound waves strike the eardrum, they cause it to vibrate. The eardrum is connected to three tiny bones: the hammer, the anvil, and the stirrup (also known by their Latin names: malleus, incus, and stapes). As the weak vibrations of the large eardrum travel

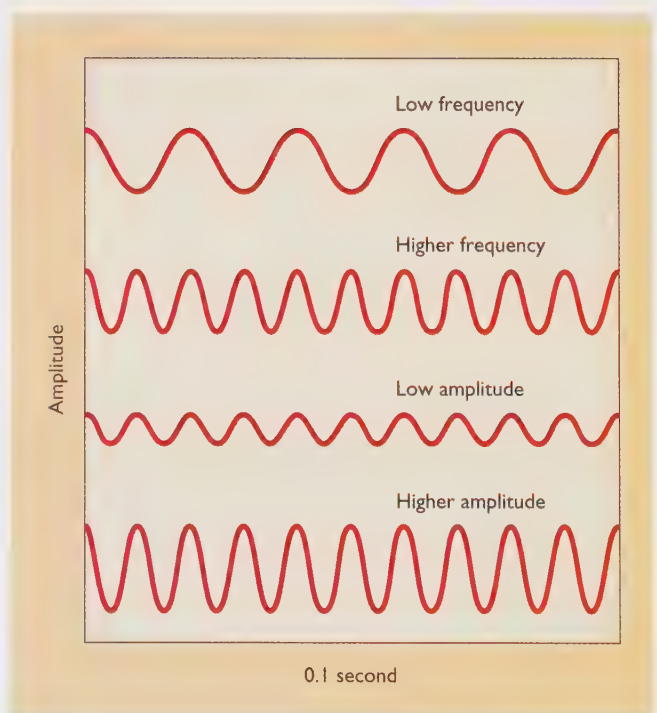


FIGURE 4.20 The period (time) between the peaks of a sound wave determines the frequency of the sound; we experience frequencies as different pitches. The vertical range, or amplitude, of a wave determines the sound's intensity and loudness.

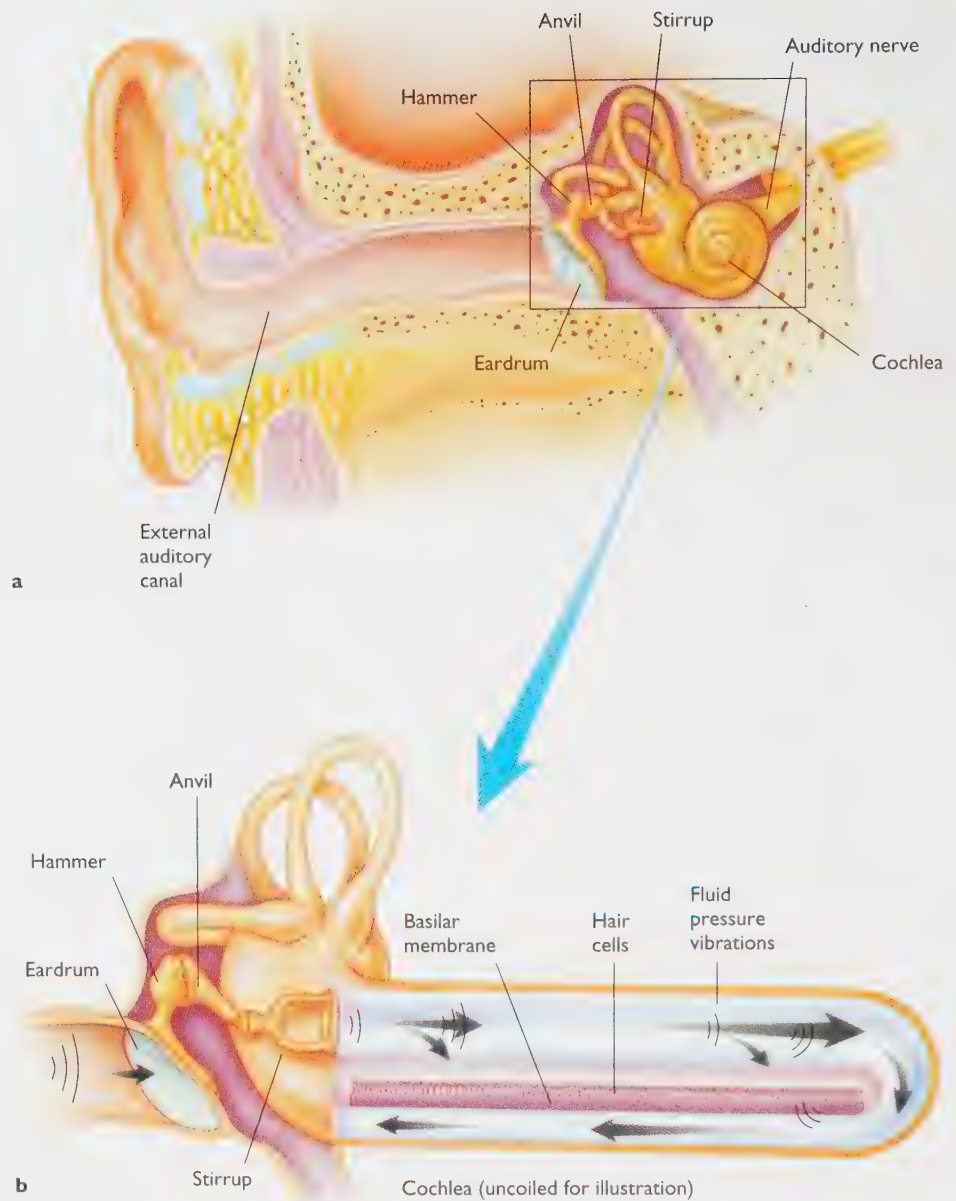


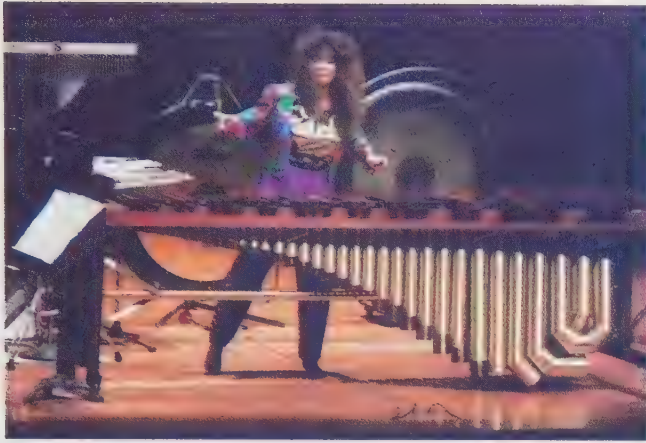
FIGURE 4.21 When sound waves strike the eardrum (a), they cause it to vibrate. The eardrum is connected to three tiny bones—the hammer, anvil, and stirrup—that convert the sound wave into a series of strong vibrations in the fluid-filled cochlea (b). Those vibrations displace the hair cells along the basilar membrane in the cochlea, which is aptly named after the Greek word for *snail*. Here, the dimensions of the cochlea have been changed to make the general principles clear.

through these bones, they are transformed into stronger vibrations of the much smaller stirrup. The stirrup in turn transmits the vibrations to the fluid-filled cochlea, where the vibrations displace hair cells along the **basilar membrane**, a thin structure within the cochlea. These hair cells, which act much like touch receptors on the skin, are connected to neurons whose axons form the auditory nerve. Impulses are transmitted along this pathway to the brain areas responsible for hearing.

A person can lose hearing in two ways. One is **conduction deafness**, which results when the bones connected to the eardrum fail to transmit sound waves properly to the cochlea. Surgery can sometimes correct conduction deafness by removing whatever is obstructing the movement of those bones. A person with conduction deafness can still hear his or her own voice, because it is conducted through the skull bones to the cochlea, bypassing the eardrum altogether. The other type of hearing loss is

nerve deafness, which results from damage to the cochlea, the hair cells, or the auditory nerve. Nerve deafness can result from heredity, from multiple sclerosis and other diseases, or from prolonged exposure to loud noises. Nerve deafness is permanent and cannot be corrected by surgery.

Hearing aids can compensate for hearing loss in most people with either conduction deafness or nerve deafness (Moore, 1989). Hearing aids merely increase the intensity of the sound, however, so they are of little help to people with severe damage to the cochlea or the auditory nerve. Many people have hearing impairments for only certain frequencies, though. For example, people with damage to certain parts of the cochlea have trouble hearing high frequencies or medium-range frequencies. Modern hearing aids can be adjusted to intensify only a certain range of sounds, so that they do not intensify sounds that are already loud enough.



Hearing is the sensing of vibrations. Evelyn Glennie, profoundly deaf since childhood, has become a famous percussionist. Although she cannot hear her music, she detects the vibrations through her stocking feet.

Pitch Perception

Adult humans can hear sound waves from about 15–20 hertz to about 15,000–20,000 Hz (cycles per second). The low frequencies are perceived as deep tones of low pitch; the high frequencies are perceived as tones of high pitch. The upper limit of hearing declines suddenly after exposure to loud noises and declines steadily as a person grows older. Thus, children can hear high-frequency sounds that adults do not.

We hear pitch by different mechanisms at different frequencies. At low frequencies (up to about 100 Hz), the *basilar membrane in the cochlea vibrates in synchrony with the sound waves; that is, it produces action potentials at the same frequency as the sound*. This is the **frequency principle**. A sound with a frequency of 50 Hz excites each hair cell along the membrane 50 times per second, thus sending 50 impulses per second to the brain.

At higher frequencies, the basilar membrane continues to vibrate in synchrony with the sound waves, but the individual hair cells cannot keep pace. (A neuron cannot fire more than about 1,000 action potentials per second, and it

cannot maintain that pace for long.) Even so, each vibration of the membrane excites at least a few hair cells, and *groups of them, volleys, respond to each vibration by producing an action potential* (Rose, Brugge, Anderson, & Hind, 1967). This is known as the **volley principle**. Thus, a tone at 1000 Hz might send impulses to the brain 1,000 times per second, even though no single neuron was firing that rapidly. Volleys are adequate for detecting most sounds of speech and music. However, even volleys cannot keep pace with frequencies approaching 5000 Hz (as compared to 4186 Hz, the highest note on a piano).

At high frequencies, sound waves of different frequencies cause vibrations at different locations along the basilar membrane. The membrane is thin and stiff near the stirrup and wide and floppy at the other end. Consequently, *high-frequency sounds cause maximum vibration near the stirrup end, and lower-frequency sounds cause maximum vibration at points farther along the membrane, thereby stimulating hair cells at different locations*. This is the **place principle**. At frequencies less than about 1000 Hz, the difference in location between one tone and another is imprecise, so the place principle does not apply at low frequencies. We do not need it at low frequencies, however, because the frequency and volley principles serve our purposes quite well. The place principle starts to become helpful beyond 1000 Hz and especially beyond 5000 Hz. In short, we detect tones up to about 1000 Hz by the frequency and volley principles, tones above 5,000 Hz by the place principle, and intermediate tones by a combination of volleys and place (Goldstein, 1989; Zwislocki, 1981). Figure 4.22 summarizes the three principles of pitch perception.

CONCEPT CHECKS

8. Suppose a mouse emits a soft, high-frequency squeak in a room full of people. Which kinds of people are least likely to hear the squeak?
9. When hair cells at one point along the basilar membrane produce 50 impulses per second, we hear a

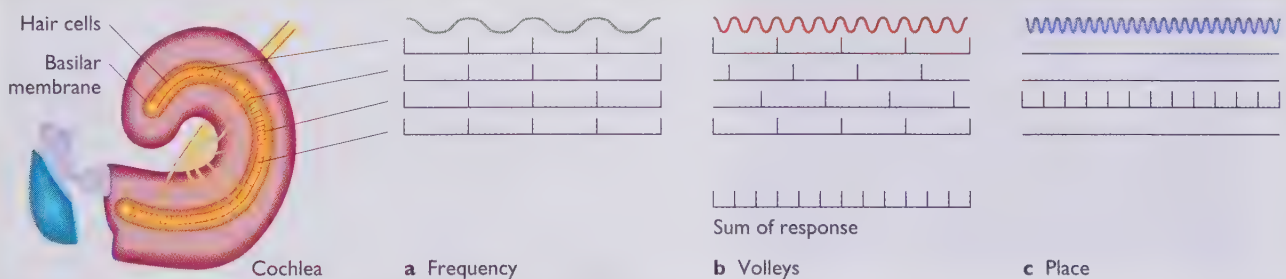


FIGURE 4.22 The auditory system responds differently to low-, medium-, and high-frequency tones. (a) At low frequencies, hair cells at many points along the basilar membrane produce impulses in synchrony with the sound waves. (b) At medium frequencies, different cells produce impulses in synchrony with different sound waves, but the group as a whole still produces one or more impulses for each wave. (c) At high frequencies, only one point along the basilar membrane vibrates; hair cells at other locations remain still.

tone at 5000 Hz. What do we hear when the same hair cells produce 100 impulses per second? (Check your answers on pages 130–131.)

Localization of Sounds

When you hear something, the stimulus is actually on the basilar membrane of your ear, but you do not experience it as such. You experience the sound as “out there,” and you can generally estimate its approximate place of origin. How do you do that?

The auditory system determines the direction of a source of sound by comparing the messages coming from the two ears. If a sound is coming from a source directly in front, the messages will arrive at the two ears at the same time and will be equal in loudness. If the sound is coming from a source on the left, however, it will arrive at the left ear slightly before it arrives at the right ear, and it will be louder in the left ear (Figure 4.23). You do not hear two sounds; you have an experience of a single sound coming from the left. A difference between the messages in the two ears indicates how far the sound source is to the left or right of center. (Someone who wears a hearing aid in just one ear may localize sounds inaccurately.)



FIGURE 4.23 The stereophonic hearing of our ears enables us to determine where a sound is coming from. The ear located closest to the sound will receive the sound waves first. A change of less than one ten-thousandth of one second can alter our perception of the location of a sound source.

The auditory system can also detect the approximate distance of a sound source. If a sound grows louder, you interpret it as coming closer. If two sounds differ in pitch, you assume the one with more high-frequency tones is closer. (Low-frequency tones carry better over a long distance than high-frequency tones do.) However, loudness and frequency tell you only the *relative* distances of sound sources; neither one provides information about the *absolute* distance. The only cue for absolute distance is the amount of reverberation (Mershon & King, 1975). In a closed room, you will first hear the sound waves that come directly from the source and then, after a delay, the waves that are reflected off the walls, floor, ceiling, and objects in the room. The more distant the source, the greater the percentage of reflected and delayed sound you will hear. When you hear many reflected sounds (echoes), you judge the source of the sound to be far away. In a noisy room, the echoes are hard to hear, so people have trouble estimating the distances of sounds; because they hear few echoes, they interpret all sounds as coming from a short distance (McMurtry & Mershon, 1985).

CONCEPT CHECKS

10. Why is it difficult to tell whether a sound is coming from directly in front of or directly behind you?
11. Suppose you are listening to a monaural (nonstereo) radio. Can the station play sounds that you will localize as coming from different directions, such as left, center, and right? Can it play sounds that you will localize as coming from different distances? Why or why not? (Check your answers on page 131.)

The Vestibular Sense

In the inner ear on each side of the head, adjacent to the structures responsible for hearing, is a structure called the *vestibule*. The **vestibular sense** that it controls *tells us the direction of tilt and amount of acceleration of the head and the position of the head with respect to gravity*. It plays a key role in posture and balance and is responsible for the sensations we experience when we are riding on a roller coaster or sitting in an airplane during takeoff.

The vestibular sense also enables us to keep our eyes fixated on a target even when our head is moving. When you walk down the street, you can keep your eyes fixated on a distant street sign, even though your head is bobbing up and down. The vestibular sense detects each head movement and controls the movement of your eyes to compensate for it.

TRY IT YOURSELF

To illustrate: Try to read this page while you are jiggling the book up and down and from side to side, keeping your head steady. Then hold the book steady and move your head up and down and from side to side. If you are like most people, you will find it much easier to read when you are moving your head



Twisting, turning, and tilting activate vestibular sensation, which reports the position of the head.

than when you are jiggling the book. That is because your vestibular sense keeps your eyes fixated on the print during head movements. People who have suffered injury to their vestibular sense report that their vision is blurry while they are walking. To read street signs or clocks, they must come to a stop and hold their head steady.

The vestibular system is composed of three semicircular canals, oriented in three separate directions, and two otolith organs (Figure 4.24b). The *semicircular canals* are lined with hair cells and filled with a jellylike substance. When the body accelerates in any direction, the jellylike substance in the corresponding semicircular canal pushes against the hair cells, which send messages to the brain. The two *otolith organs* shown in Figure 4.24b also contain hair cells (Figure 4.24c), which lie next to the *otoliths* (calcium carbonate particles). Depending on which way the head tilts, the particles move about in the direction of gravitational pull and excite different sets of hair cells. The otolith organs report the direction from which gravity is pulling and tell us which way is “up.”

What happens if the otoliths fail to provide reliable information? For astronauts in the zero-gravity environment of outer space, the otoliths provide no useful information about up or down; indeed, the up-down dimension is almost meaningless to them. Instead, they learn to rely entirely on visual signals, such as closeness to one object or distance from another (Lackner, 1993).

The Cutaneous Senses

What we commonly think of as the sense of touch actually consists of several partly independent senses: pressure on the skin, warmth, cold, pain, vibration, movement across the skin, and stretch of the skin. These sensations depend on several kinds of receptors in the skin, as Figure 4.25 shows (Iggo & Andres, 1982). A pinch on the skin feels different from a tickle, and both feel different from a cut or a burn, because each of these stimuli excites different recep-

tors. Collectively, these sensations are known as the **cutaneous senses**, meaning the *skin senses*. Although they are most prominent in the skin, we have some of the same receptors in our internal organs as well, enabling us to feel internal pain, pressure, or temperature changes. Therefore, the cutaneous senses are sometimes known by the broader term *somatosensory system*, meaning *body-sensory system*.

On the fingertips, the lips, and other highly sensitive areas of skin, the receptors are densely packed, and each receptor detects stimulation in only a small area of the skin. On the back and other less sensitive areas, the receptors are scattered more widely, and each one is responsible for detecting stimulation over a large surface. Similarly, much

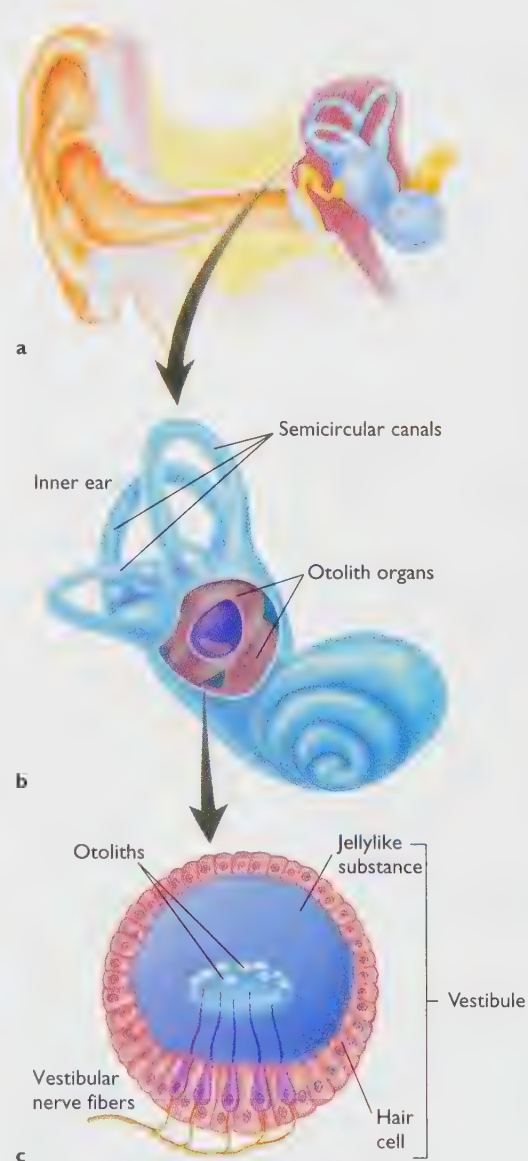


FIGURE 4.24 (a) Location of and (b) structures of the vestibule. (c) Moving your head or body displaces hair cells that report the tilt of your head and the direction and acceleration of movement.

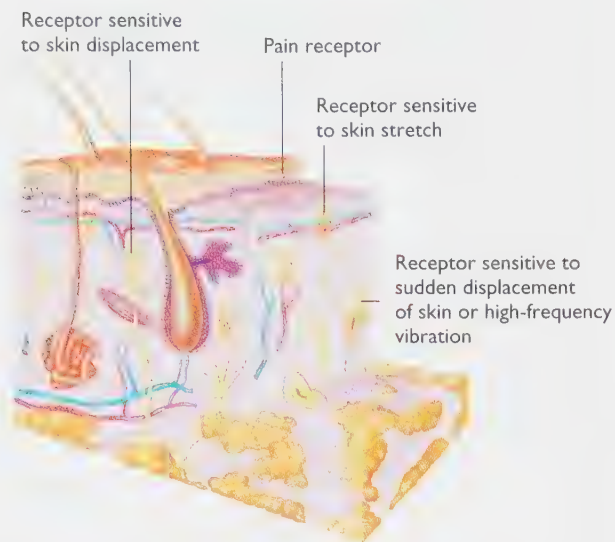


FIGURE 4.25 Cutaneous sensation is the product of many kinds of receptors, each sensitive to a particular kind of information.

more of the parietal lobe is devoted to sensation from the lips and fingers than from the less sensitive areas.

If you ask someone to place an everyday object in your hand (such as a pencil or a spoon), without showing it to you or telling you what it is, you can probably identify the object just by feeling it. You will probably fail to identify a more complicated or less familiar object, however, such as a scale model of the Eiffel Tower, even though you could identify it easily by sight. With extensive practice, you could improve your ability to identify objects by touch. Subjects in one study felt raised-line drawings, like those in Figure 4.26, without seeing them. Some of the subjects were sighted; some were blind since birth; and some had become blind later in life. Most sighted people found it very difficult to identify what the drawings represented, presumably because they had little practice at paying close attention to touch. People blind since birth also had little success with this task, but for a different reason: A raised-line drawing of an umbrella or similar object makes little sense to someone who has never seen a drawing of the same object. In contrast, people who had lost their vision later in life were able to identify many of the objects (Heller, 1989). They had the advantage of previous experience with visual drawings, plus years of practice in paying close attention to touch.

Pain

Pain receptors are simple, bare nerve endings that send messages to the spinal cord. The experience of pain, however, is a complicated mixture of sensation (the information about tissue damage) and emotion (the unpleasant reaction). The sensory and emotional qualities are governed by different parts of the brain (Craig, Bushnell, Zhang, & Blomqvist, 1994; Fernandez & Turk, 1992), so the emotional reaction can be either much stronger or weaker than the sensation itself.

The Gate Theory of Pain You visit a physician because of severe pain, but as soon as the physician tells you the problem is minor and nothing to worry about, the pain starts to subside. Have you ever had such an experience? Pain can increase or decrease greatly just because of people's expectations. Recall the term *placebo* from Chapter 2: A placebo is a drug or other procedure that has no pharmacological properties, other than those that result from people's expectations; researchers ordinarily give placebos to control groups. Placebos can produce such large effects on pain that they become a research topic on their own. Consider the following: In one experiment, college students were given a smelly brownish liquid on the index finger of one hand. It was in fact just a placebo, but they were told that it was a painkiller. Then the students were given a painful pinch stimulus to that finger and the index finger of the other hand. They consistently reported the pain on the finger with the placebo to be less intense than the pain on the other finger (Montgomery & Kirsch, 1996). How placebos work is far from clear, but these results eliminate mere relaxation, which would presumably reduce pain on both hands equally.

Think about the various times you have had cuts or other minor injuries. Under most conditions, you have probably found that you can relieve the pain by gently massaging the skin near the cut, or by applying either cold packs



FIGURE 4.26 Sighted and blind subjects felt these raised-line drawings and tried to identify what they represented. Most sighted subjects and subjects blind since birth found the task very difficult and seldom answered correctly. Subjects who had become blind later in life performed much better (Heller, 1989).

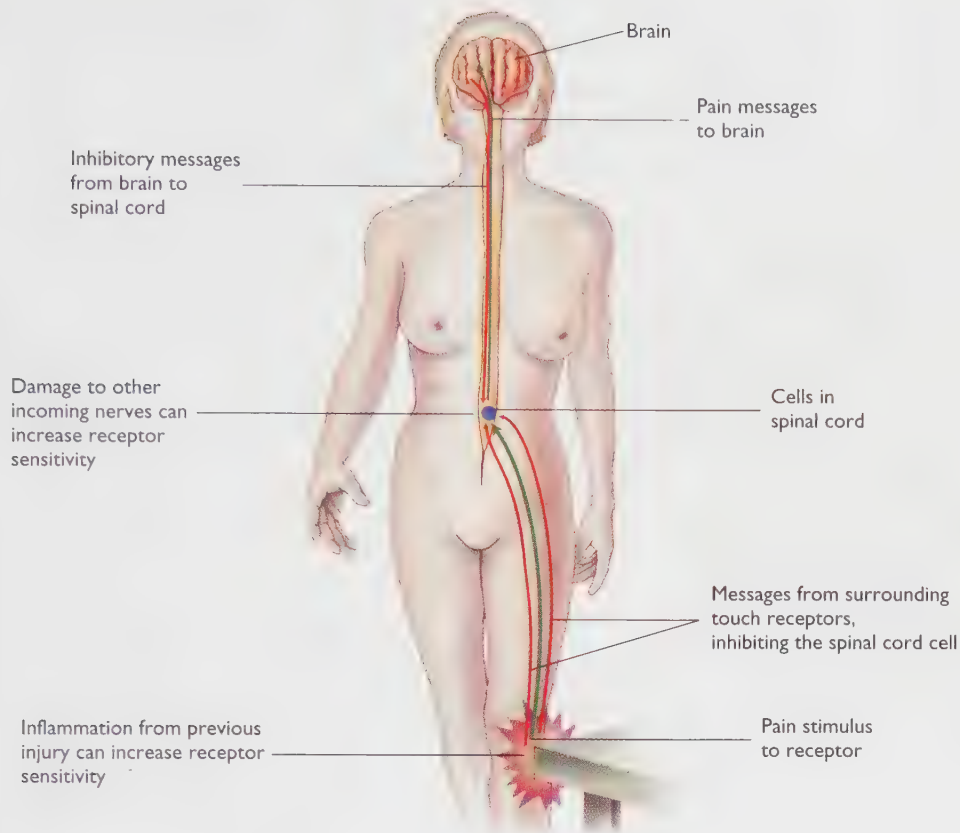


FIGURE 4.27 Pain messages from the skin are relayed from spinal cord cells to the brain. According to the gate theory of pain, those spinal cord cells serve as a gate that can block or enhance the signal. The proposed neural circuitry is simplified in this diagram. Green lines indicate axons with excitatory inputs; red lines indicate axons with inhibitory inputs.

or hot packs (Rollman, 1991). However, when you have had a sunburn, you probably found that even a light touch anywhere on the sunburned area, or stepping into a hot shower, became excruciatingly painful (Devor, 1996). A variety of processes can either increase or decrease the pain caused by a particular injury.

Because of observations such as these, Ronald Melzack and P. D. Wall (1965) proposed the **gate theory** of pain, the idea that *pain messages must pass through a gate, presumably in the spinal cord, that can block these messages*. For example, rubbing the surrounding skin sends inhibitory messages to the spinal cord, closing the pain gates. Various pleasant or distracting brain activities can also send inhibitory messages. Later researchers found that the gate can also enhance the pain messages; for example, damage to some of the incoming axons can induce increased sensitivity of the spinal cord neurons, so that weak stimulation can seem very painful. In short, the activities of the rest of the nervous system can facilitate or inhibit the transmission of pain messages (Figure 4.27).

Mechanisms of Decreasing Pain Pain alerts us to an injury. A small percentage of people are completely insensitive to pain. They can burn themselves by picking up hot objects; they can scald their tongues on hot coffee; they can cut themselves without realizing it; they can bite their tongues hard, possibly even biting off the tip; they can sit in

a single position for hours without growing uncomfortable, thereby damaging their bones and tendons (Comings & Amromin, 1974).

Although pain messages serve an essential function by alerting us to danger and damage, a prolonged, intense pain message is unnecessary and sometimes disruptive. One way to reduce the sensation of pain is to provide some distraction. In terms of the gate theory, a distraction closes the pain gate. For example, surgery patients in a room with a pleasant view complain less about pain, take less painkilling medicine, and recover faster than do patients in a windowless room or a room with a poor view (Ulrich, 1984).

*Pain messages in the nervous system release a neurotransmitter called **substance P**, as well as several others.* The neurons that have receptors for substance P are essential for pain; after injections of chemicals that destroy those neurons, laboratory animals have become almost unresponsive to many kinds of painful stimuli (Mantyh et al., 1997). Another set of synapses release **endorphins**, *neurotransmitters that inhibit the release of substance P and thereby decrease pain sensations* (Reichling, Kwiat, & Basbaum, 1988; Terman, Shavitt, Lewis, Cannon, & Liebeskind, 1984). (See Figure 4.28.) The term *endorphin* is a combination of the terms *endogenous* (self-produced) and *morphine*. The drug morphine has long been known for its exceptional ability to inhibit pain; it does so by stim-

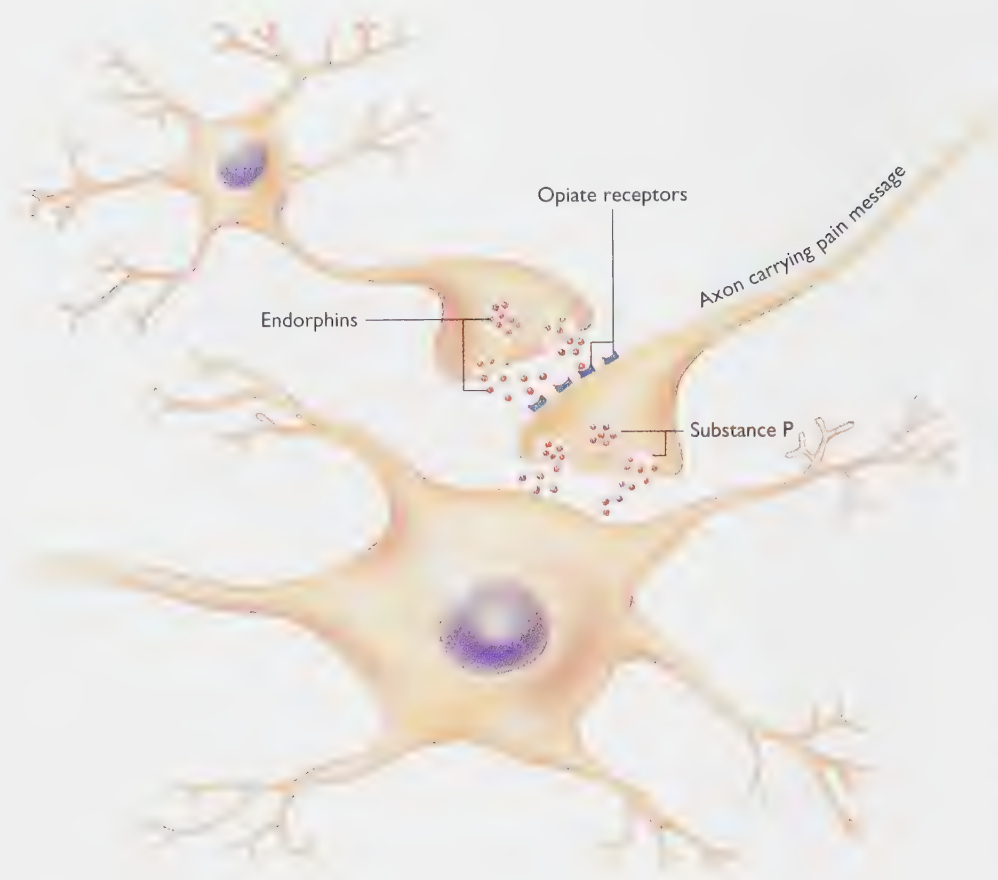


FIGURE 4.28 Substance P is the neurotransmitter most responsible for pain sensations. Endorphins are neurotransmitters that block the release of substance P, thereby decreasing pain sensations. Opiates decrease pain by mimicking the effects of endorphins.

ulating the endorphin synapses in the brain. A variety of nondrug stimuli can also release endorphins. Under some circumstances, a painful stimulus itself releases endorphins so that exposure to one painful stimulus decreases sensitivity to the next painful stimulus (Terman & Liebeskind, 1986). Pleasant stimuli can also release endorphins. (That effect may help to explain why a pleasant view helps to ease postsurgical pain.) In short, endorphins are a powerful method, perhaps the main method, of closing pain “gates.”

Paradoxically, another method of decreasing pain begins by inducing it. The **chemical capsaicin** stimulates receptors that respond to painful heat (Caterina et al., 1997), and thereby *causes the release of substance P*. Injecting capsaicin or rubbing it on the skin produces a temporary burning or stinging sensation (Karrer & Bartoshuk, 1991; Yarsh, Farb, Leeman, & Jessell, 1979). However, because capsaicin releases substance P faster than the neurons can resynthesize it, after the burning sensation subsides, the result is a fairly long-lasting decrease in pain sensitivity. Several skin creams intended for the relief of aching muscles contain capsaicin. (Don’t rub them on right before you go to bed. They produce a burning sensation before they relieve the muscle pain.)

Jalapenos and other hot peppers contain capsaicin. The reason that they taste hot is that their capsaicin releases

enough substance P from the tongue to cause a stinging, hot sensation.

CONCEPT CHECKS

- 12.** Naloxone, a drug used as an antidote for an overdose of morphine, is known to block the endorphin synapses. How could we use naloxone to determine whether a pleasant stimulus releases endorphins?
- 13.** Psychologist Linda Bartoshuk recommends candies containing moderate amounts of jalapeño peppers as a treatment for people with pain in the mouth. Why? (Check your answers on page 131.)

The Chemical Senses: Taste and Smell

Most textbooks on sensation concentrate on vision and hearing; some ignore taste and smell or include them in a chapter titled “The Other Senses” or even “The Minor Senses.” For most of the animal kingdom, however, these senses are not so minor. If rats or raccoons wrote sensation textbooks, they would probably devote as much coverage to taste and smell as they would to hearing, and they would list vision as one of the

“minor senses.” Many invertebrates have no vision or hearing at all; they survive with only chemical senses and touch.

Taste

Vision and hearing enable us to find food and water, to avoid danger, to keep our balance, and to find suitable mates. The sense of **taste**, which *detects chemicals on the tongue*, serves just one function: It tells us what to eat and drink.

The **taste receptors** are in the **taste buds**, located in the folds on the surface of the tongue, almost exclusively along the outside edge of the tongue in adults (Figure 4.29).

TRY IT YOURSELF

(Children's taste buds are more widely scattered.) Try this demonstration (based on Bartoshuk, 1991): Soak something small (the tip of a paper napkin will do) in sugar water, salt water, or vinegar. Then dab it onto the center of your tongue. You will taste nothing. Then slowly move the soaked substance toward the side of the tongue. Suddenly, you taste it.

If you go in the other direction (first touching the side of the tongue and then moving toward the center), you will continue to taste the substance even when it reaches the center of your tongue. The explanation is not that you suddenly grew new taste buds. Rather, your taste buds provide no information about location. Once you have stimulated the taste buds near the edge of the tongue, you will continue tasting the substance, but the taste receptors do not tell you *where* you are tasting the substance. If you now stimulate *touch* receptors elsewhere on your tongue, your brain interprets the taste perception as coming from the spot you are touching, even though the taste sensation is in fact coming from a different location (the side of your tongue).

Different Types of Taste Receptors Researchers now have a reasonably clear understanding of the structures of the taste receptors and how they work (Margolskee, 1993). Actually, long before neuroscientists began characterizing taste receptors, behavioral researchers had solid evidence that different tastes depend on different kinds of receptors—probably four kinds, representing sweet, sour, salty, and bitter. That conclusion rested partly on demonstrations that people can match most tastes by mixing sweet, sour, salty, and bitter substances (Schiffman & Erickson, 1971). The conclusion was also supported by evidence that certain procedures can affect one taste without affecting others, presumably by acting on only one type of receptor. For example, did you ever drink a glass of orange juice just after brushing your teeth? How can something that ordinarily tastes so good suddenly taste so bad? The reason is that most toothpastes contain sodium lauryl sulfate, a chemical that weakens sweet tastes and intensifies sour and bitter tastes (Schiffman, 1983).

People from different parts of the world have different taste preferences. Contrast, for example, Greek cuisine, Mexican cuisine, and Chinese cuisine. Do the different cultural food preferences relate in any way to differences in people's sense of taste? Evidently not. People are about the same throughout the world in their ability to taste various substances (e.g., Laing et al., 1993). Their food preferences depend mainly on what is familiar to them.

Olfaction

Olfaction is the *sense of smell*. The olfactory receptors, located on the mucous membrane in the rear air passages of the nose (Figure 4.30b), detect the presence of certain air-

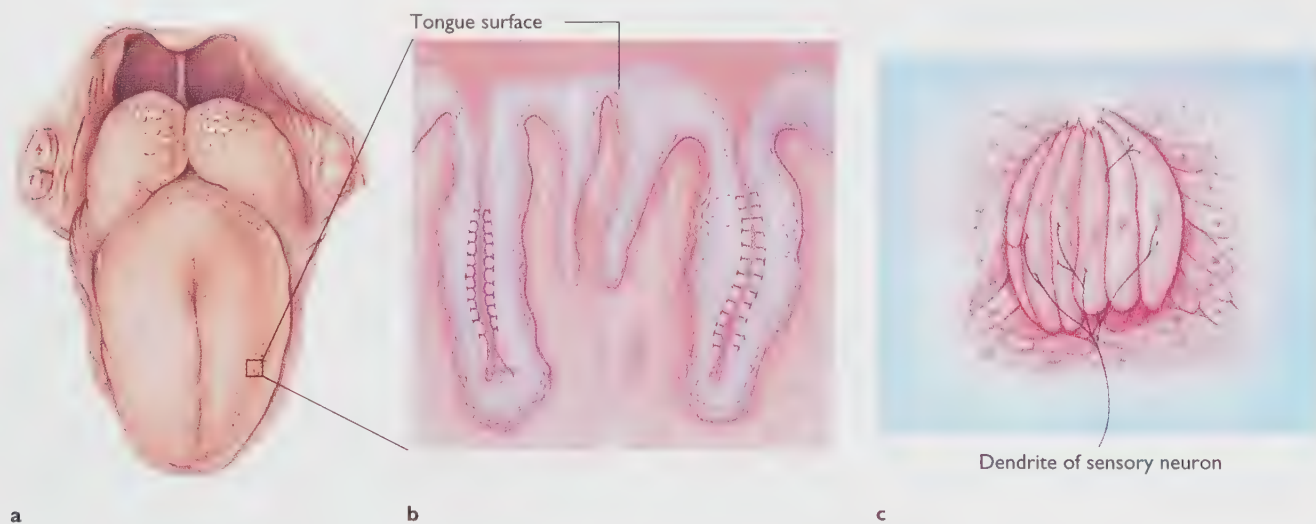


FIGURE 4.29 (a) The tongue, a powerful muscle used for speaking and eating: Taste buds, which react to chemicals dissolved in saliva, are located along the edge of the tongue in adult humans but are more widely distributed in children. (b) A cross section through part of the surface of the tongue, showing taste buds. (c) A cross section of one taste bud. Each taste bud has about 50 receptor cells within it.

borne molecules. Chemically, these receptors are much like synaptic receptors, except that they are stimulated by chemicals from the environment instead of chemicals released by other neurons. The axons of the olfactory receptors form the olfactory tract, which extends to the olfactory bulbs at the base of the brain.

How many kinds of olfactory receptors do we have? Investigators have long known that color vision depends on three kinds of receptors; they have also long known approximately how many receptor types we have for hearing, touch, and taste. However, until 1991 researchers had virtually no idea how many types of receptors existed for olfaction.

In principle, researchers can determine the number of receptor types through behavioral data, without chemically isolating the receptors. With color vision, for example, the researchers of the 1800s established that people can mix three colors of light in various amounts to match any other color. Therefore, even before the technology existed to examine the cones in the retina, researchers had reason to believe that the retina has three kinds of cones. In taste, the fact that we can mix sweet, sour, salty, and bitter substances to match almost any other taste implies that the tongue has

perhaps only four kinds of taste receptors. Regarding olfaction, however, no one knew how many kinds of receptors to expect. Can people match all possible odors by mixing appropriate amounts of three, four, seven, or ten “primary” odors? or do they need 50, 100, or what? No researcher had ever demonstrated that it was possible to match all the possible odors by mixing any number of primaries.

Perhaps it is just as well that no one spent a lifetime investigating this issue. In 1991, Linda Buck and Richard Axel used the latest biochemical technology to demonstrate that the nose has at least a hundred types of olfactory receptors, and probably several hundred (Buck & Axel, 1991). In most cases, a given receptor responds well to only a limited family of odorant molecules (Ressler, Sullivan, & Buck, 1994). We do not yet know exactly how the brain makes sense of hundreds of channels of olfactory information (Figure 4.31). What we do know is that our olfactory system is able to detect and discriminate among an enormous number of possible molecules. When perfume chemists synthesize a brand-new molecule, people do not need to evolve a new receptor to detect it; we can detect any chemical with some combination of the receptors we already have.

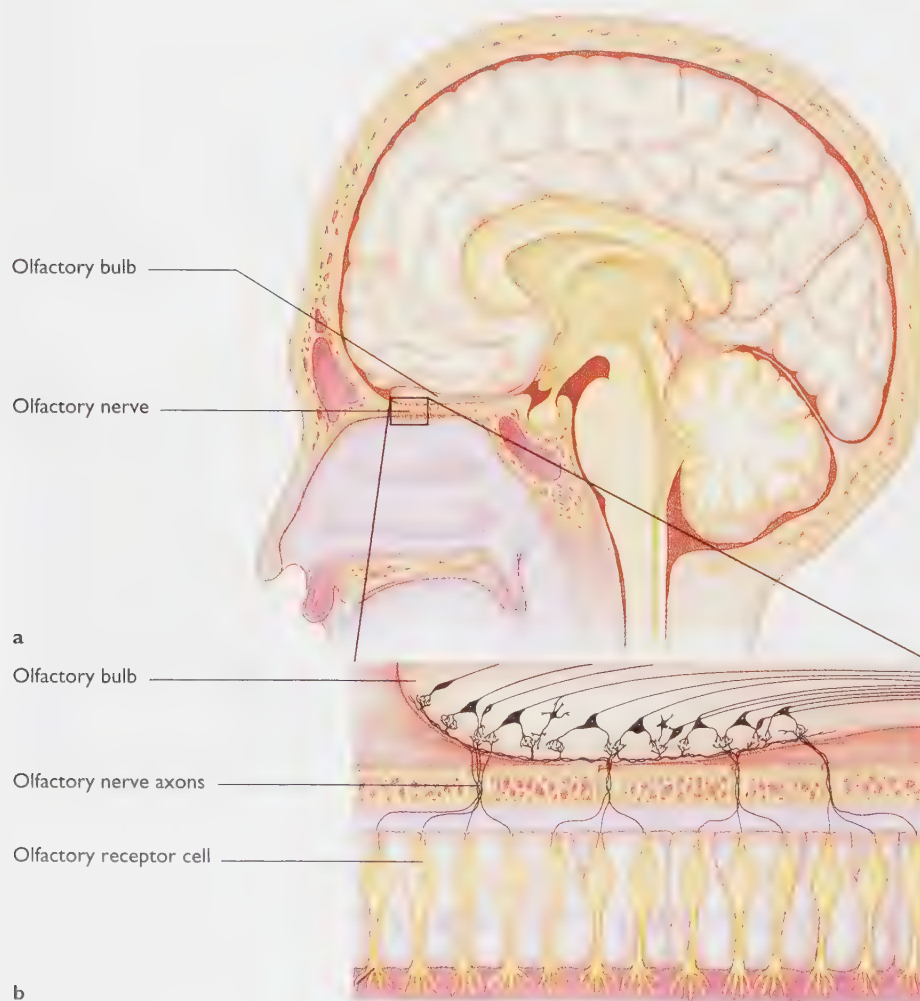


FIGURE 4.30 The olfactory receptor cells lining the nasal cavity send information to the olfactory bulb in the brain. There are at least 100 types of receptors with specialized responses to airborne chemicals.

Still, humans' olfactory abilities are unimpressive compared to most other mammals. A single olfactory receptor is probably about as sensitive in humans as it is in any other species, but most other mammals have a greater number of receptors. For example, trained dogs can track a person's olfactory trail across fields and through woods. It is difficult (though humorous) to imagine a human even attempting such a task. People can improve their olfactory discrimination by practice, however. For example, professional wine tasters become highly skilled at discriminating one wine odor from another (Bende & Nordin, 1997).

Olfaction is particularly important for food selection. Neurons in the prefrontal cortex receive both taste and olfactory information, producing the combined sensation that we call *flavor*. These cells also receive input that indicates hunger, and they respond vigorously to the flavor of a food only when the individual is hungry (Rolls, 1997).

Olfaction also serves social functions, especially in nonhuman mammals that identify one another by means of **pheromones**, *odorous chemicals they release into the environment*. Pheromones act primarily on the vomeronasal organ, a set of receptors near, but separate from, the standard olfactory receptors (Monti-Bloch, Jennings-White, Dolberg, & Berliner, 1994). In nearly all nonhuman mammals, the males rely on pheromones to distinguish sexually receptive females from unreceptive females.

Humans prefer *not* to recognize one another by smell. The deodorant and perfume industries exist for the sole purpose of removing and covering up human odors. But perhaps we respond to pheromones anyway, at least under certain conditions. For example, young women who are in frequent contact, such as roommates in a college dormitory, tend to synchronize their menstrual cycles, probably as a result of pheromones they secrete (McClintock, 1971). One study examined women in Bedouin Arab families. The advantages of studying that culture are that the mother and



Professional deodorant tester: That's a career option you probably never even considered. Secret, Ban, Arrid—as their names suggest, U.S. industries spend millions of dollars developing and promoting deodorants and antiperspirants so that we can secretly banish sweat and have arid armpits.

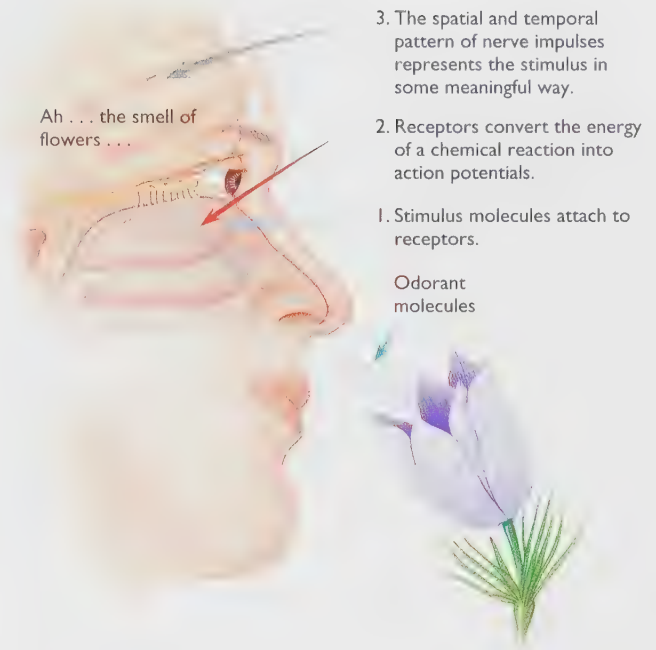


FIGURE 4.31 Olfaction, like any other sensory system, converts physical energy into a complex pattern of brain activity.

sisters within a family have extensive daily contact, unmarried women have almost no contact with men, and virtually none of the unmarried women and few of the married women use oral contraceptives. Thus pheromones have a maximum opportunity to show their effects in Bedouin culture. The results showed that the women in most families were at least partly synchronized; they might not begin to menstruate on exactly the same day, but they were close (Weller & Weller, 1997).

THE MESSAGE

Sensory Systems

The world as experienced by a bat (which can hear frequencies of 100,000 Hz) or a dog (which can discriminate among odors that you and I would never notice) or a mouse (which depends on its whiskers when exploring the world) is in many ways a different world from the one that people experience. The function of our senses is not to tell us about everything in the world, but to alert us to the information we are most likely to use, given our way of life.

SUMMARY

★ **Pitch.** At low frequencies of sound, we identify pitch by the frequency of vibrations on the basilar membrane. At intermediate frequencies, we identify pitch by volleys of responses from a variety of neurons. At high frequencies, we identify pitch by the area of the basilar membrane that vibrates most strongly. (page 121)

★ **Localizing sounds.** We localize the source of a sound by detecting differences in the time and loudness of the sounds our two ears receive. We localize the distance of a sound source mostly by the amount of reverberation, or echoes, following the main sound. (page 122)

★ **Vestibular system.** The vestibular system tells us about the movement of the head and its position with respect to gravity. The vestibular system enables us to keep our eyes fixated on an object while the rest of our body is in motion. (page 122)

★ **Cutaneous receptors.** We experience many types of sensation on the skin, each dependent on different receptors. The fingertips, lips, and face have especially rich supplies of such receptors. (page 123)

★ **Pain.** The experience of pain can be greatly inhibited or enhanced by other simultaneous experiences, including touch to surrounding skin or the person's expectations. Pain depends largely on stimulation of neurons that are sensitive to the neurotransmitter substance P, which can be inhibited by endorphins. (page 124)

★ **Taste receptors.** Even before researchers identified the taste receptors, they knew that there must be at least four kinds, because certain procedures affect one taste quality (such as sweetness) without affecting the others. An adult human has taste receptors only along the edges of the tongue. (page 127)

★ **Olfactory receptors.** The olfactory system—the sense of smell—depends on at least a hundred types of receptors, each with its own special sensitivity. Olfaction is important for many behaviors, including food selection and (especially in mammals) identification of potential mates. (page 128)

Suggestions for Further Reading

- Heller, M. A., & Schiff, W. (1991). *The psychology of touch*. Hillsdale, NJ: Erlbaum. Describes research on touch and how people learn to use it for Braille reading and other functions.
- McLaughlin, S., & Margolskee, R. F. (1994). The sense of taste. *American Scientist*, 82, 538–545. Highly informative discussion of the physiology and psychology of taste.
- Zwislocki, J. J. (1981). Sound analysis in the ear: A history of discoveries. *American Scientist*, 69, 184–192. Review of research on the mechanisms of hearing.

Terms

- sound waves** vibrations of the air or of another medium (page 119)
- hertz (Hz)** a unit of frequency representing one cycle (vibration) per second (page 119)
- pitch** a perception closely related to the frequency of sound waves (page 119)
- loudness** a perception that depends on the amplitude of a sound wave (page 119)
- cochlea** the snail-shaped, fluid-filled structure that contains the receptors for hearing (page 119)

basilar membrane a thin structure in the cochlea that vibrates after sound waves strike the eardrum (page 120)

conduction deafness hearing loss that results when the bones connected to the eardrum fail to transmit sound waves properly to the cochlea (page 120)

nerve deafness hearing loss that results from damage to the cochlea, the hair cells, or the auditory nerve (page 120)

frequency principle identification of pitch by the frequency of action potentials in neurons along the basilar membrane of the cochlea, synchronized with the frequency of sound waves (page 121)

volley principle identification of pitch by groups of hair cells responding to each vibration by producing an action potential (page 121)

place principle identification of pitch by determining which auditory neurons, coming from which part of the basilar membrane, are most active (page 121)

vestibular sense a specialized sense that detects the direction of tilt and amount of acceleration of the head and the position of the head with respect to gravity (page 122)

cutaneous senses the skin senses, including pressure on the skin, warmth, cold, pain, vibration, movement across the skin, and stretch of the skin (page 123)

gate theory the proposal that pain messages must pass through a gate, probably in the spinal cord, that can block these messages (page 125)

substance P a neurotransmitter responsible for much of the transmission of pain information in the nervous system (page 125)

endorphin any of the neurotransmitters that decrease the perception of pain (page 125)

capsaicin a chemical that stimulates the release of substance P (page 126)

taste the sensory system that responds to chemicals on the tongue (page 127)

taste bud the site of the taste receptors, located in one of the folds on the surface of the tongue (page 127)

olfaction the sense of smell; the detection of chemicals in contact with the membranes inside the nose (page 127)

pheromone an odorous chemical, released by an animal, that changes how other members of the species respond to that animal socially (page 129)

Answers to Concept Checks

- Obviously, the people farthest from the mouse are least likely to hear it. In addition, older people would be less likely than young people are to hear the squeak, because the ability to hear high frequencies declines in old age. Another group unlikely to hear the squeak are those who have had repeated exposure to loud noises. For this reason, you should avoid attending loud rock concerts, and listening to recorded music played at a loud volume, especially when you listen on a Walkman or similar device. You could damage your hearing in the long run, even if you do not realize it now. (page 121)
- We still hear a tone at 5000 Hz, but it is louder than before. For high-frequency tones, the pitch we hear depends on which hair cells are most active, not how many impulses per second they fire. (page 121)

10. We localize sounds by comparing the input into the left ear with the input into the right ear. If a sound comes from straight ahead or from straight behind us (or from straight above or below), the input into the left ear will be identical with the input into the right ear. (page 122)
11. Various sounds from the radio cannot seem to come from different directions, because your localization of the direction of a sound depends on a comparison between the responses of the two ears. However, the radio can play sounds that seem to come from different distances, because distance localization does not depend on a difference between the ears. It depends on the amount of reverberation, loudness, and high-frequency tones, all of which can be varied with a single speaker. Consequently, the radio can easily give an impression of people walking toward you or away from you, but not of people walking left to right or right to left. (page 122)
12. First determine how much the pleasant stimulus decreases the experience of pain for several people. Then give half of them naloxone and half of them a placebo. Again measure how much the pleasant stimulus decreases the pain. If the pleasant stimulus decreases pain by releasing endorphins, then naloxone should impair its painkilling effects. (page 126)
13. The capsaicin in the jalapeño peppers will release substance P faster than it can be resynthesized, thus decreasing the later sensitivity to pain in the mouth. (page 126)

Web Resources

HyperText Psychology—SENSES/Taste

www.science.wayne.edu/~wpoff/cor/sen/taste.html

Learn how we smell odors, what things cause smells, the types of smells, and how the sense of smell can be damaged.

The Interpretation of Sensory Information

What is the relationship between the real world and the way we perceive it?

Why are we sometimes wrong about what we think we see?

No doubt you have heard people say that “a picture is worth a thousand words.” If so, what is one one-thousandth of a picture worth? one word? Ordinarily, one one-thousandth of a picture is worth nothing. (Of course, I grant, “nothing” is *one word!*)

Printed photographs, such as the one on page 129, are composed of a great many dots. Ordinarily, you will be aware of only the overall patterns and objects, but if you magnify a photo, as in Figure 4.32, you can see the individual dots. Although one dot by itself tells us almost nothing, the pattern of dots as a whole constitutes a meaningful picture.

Actually, our vision is like this all the time. Your retina is composed of about 126 million rods and cones, each of



FIGURE 4.32 Although this photograph is composed entirely of dots, we see objects and patterns. The principles at work in our perception of this photograph are at work in all our perceptions.

which sees one dot of the visual field. What you perceive is not dots, however, but lines, curves, and complex objects. In a variety of ways, your nervous system starts with an enormous array of details and extracts the meaningful information.

Perception of Minimal Stimuli

Some of the very earliest researchers in psychology addressed the question, “What is the weakest sound, the weakest light, the weakest touch, and so forth, that a person can detect?” The early psychologists assumed that this question would be easy to answer and therefore a good starting point for further research. As is often the case, however, this apparently simple question had a complicated answer: What you can detect varies from one occasion to another, and even stimuli that you say you did not see or hear can slightly influence your behavior.

Sensory Thresholds and Signal Detection

In a typical experiment to determine the threshold of hearing—that is, the minimum intensity at which humans can detect sound—subjects are presented with tones of varying intensity in random order, including some trials with no tone at all. During each trial the subjects are asked to say whether or not they heard a tone. Figure 4.33 presents typical results. Notice that people are more likely to report stronger tones than weaker tones, but they occasionally report hearing a tone even when there was no tone at all. No sharp dividing line separates sounds that people can hear from sounds they cannot.

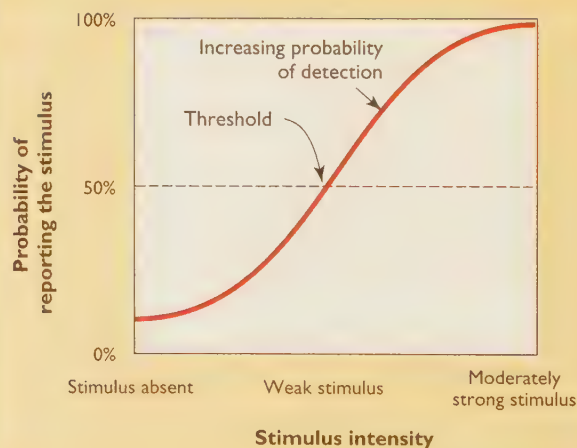


FIGURE 4.33 Typical results of an experiment to measure a sensory threshold. There is no sharp boundary between stimuli that you can perceive and stimuli that you cannot perceive.

Because perception researchers find no sharp dividing line between detectable and nondetectable stimuli, they define a **sensory threshold** as the *intensity at which a given individual can detect a stimulus 50% of the time*. Note, however, that an individual will sometimes detect stimuli below the threshold or fail to detect stimuli above the threshold.

An individual's sensory threshold can change as a result of dark adaptation. For example, if you have been outdoors on a bright, sunny day and you now walk into a movie theater, you will have trouble seeing the seats at first. After a few minutes, your threshold drops (that is, your sensitivity increases) and you can see the seats well. If you stayed in the dark theater after the movie ended, your ability to detect dim light would increase still further. The *sensory threshold at the time of maximum dark adaptation* is called the **absolute threshold**.

When people try to detect weak stimuli, they can be correct in two ways: reporting a stimulus when it is present (a "hit") and reporting no stimulus when it is absent (a "correct rejection"). They can also be wrong in two ways: They can fail to detect a stimulus (a "miss"), or they can say they detected a stimulus when none was present (a "false alarm"). Figure 4.34 outlines these possibilities.

Signal-detection theory is the *study of people's tendencies to make hits, correct rejections, misses, and false alarms* (Green & Swets, 1966). (Psychologists borrowed signal-detection theory from engineering, where this system is applied to such matters as detecting radio signals in the presence of interfering noise.) In signal-detection studies, we compare responses for stimulus-present and stimulus-absent trials. For example, suppose that someone reports a stimulus present on 80% of the trials when we present it. That statistic tells us very little by itself, unless we know how often the person said it was present on trials when it was not. If he or she also said "present" on 80% of trials with no stimulus, then the person was not actually detecting anything at all—just guessing "yes" most of the time. We would be more impressed if the person seldom reported the stimulus present when it was absent.

In a signal-detection experiment, people's responses depend on their willingness to risk a miss or a false alarm. (When in doubt, they have to risk one or the other.) Suppose you are the subject and I tell you that you will receive a 10-cent reward every time you correctly report that a light is present, but you will be fined one cent for saying "yes" if a light was not present. The lights are extremely faint, and sometimes you are not sure whether you saw one or not. When you are not sure, you will probably guess "yes," and the results will resemble those in Figure 4.35a. Then, I change the rules: You will receive a 1-cent reward for correctly reporting the presence of a light, but you will suffer a 10-cent penalty and an electric shock if you report a light when none was present. Now, you will say yes only if you are certain you saw a light, and the results will look like those in Figure 4.35b. In short, people's answers depend on the instructions they receive and the strategies they use, not just what their senses tell them.

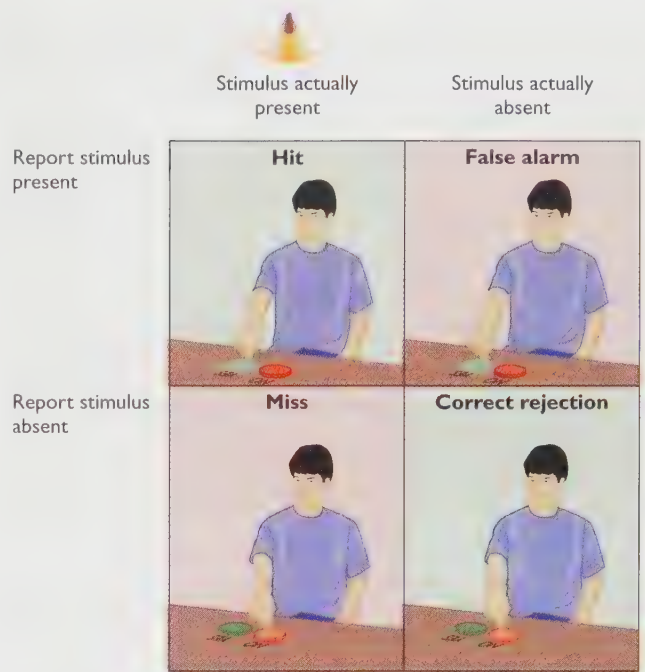
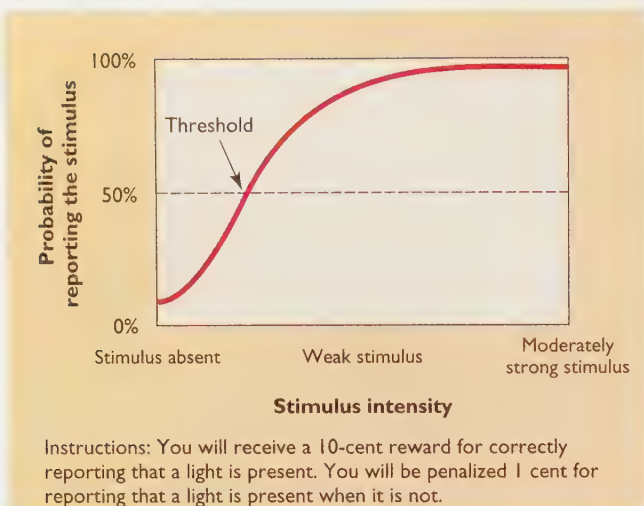


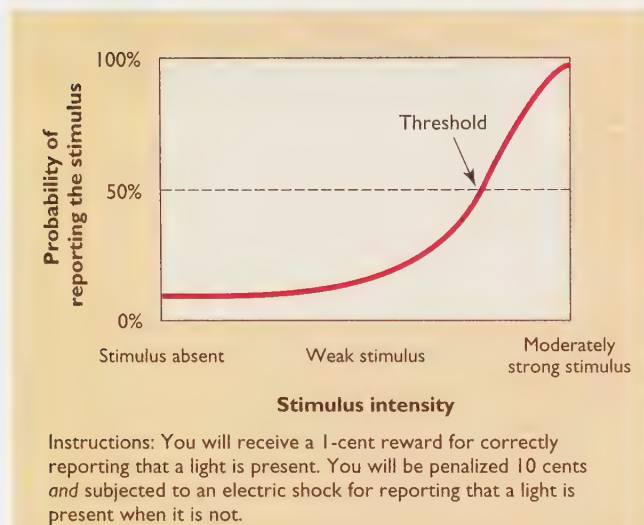
FIGURE 4.34 People can make two kinds of correct judgments (green) and two kinds of errors (red). Someone who too readily reports the stimulus present would get many hits, but also many false alarms.

People can become cautious about false alarms for other reasons, too. In one experiment, participants were asked to try to read words that were flashed on a screen for just a split second. They performed well when ordinary words like *river* or *peach* were shown. For emotionally loaded words like *penis* or *bitch*, however, they generally said they were not sure what they saw. Psychologists have suggested several possible explanations for such results (e.g., Blum & Barbour, 1979); one likely possibility is that subjects hesitate to blurt out an emotionally charged word unless they are certain they are right.

The signal-detection approach is useful in many settings that are remote from a vision or hearing laboratory. For example, if someone uses a personality test to identify people suffering from depression, it is important to evaluate both the number and kinds of errors that the test makes. A "miss" would be a failure to identify a depressed person. A "false alarm" would be calling someone depressed who in fact is not. Depending on the consequences of failing to treat someone who needs help or of providing treatment to someone who doesn't need it, we might be more willing to make one kind of mistake than the other. The legal system is also a signal-detection situation. When we evaluate the suitability of any kind of evidence—DNA matching, lie detector tests, eyewitness testimony, and so forth—the key questions are how often this kind of evidence would label a guilty person as innocent (a miss) and how often it might lead to the conviction of an innocent person (a false alarm). In the legal system, most people are much more willing to accept misses than false alarms. (That is, we reject any evidence that is likely to convict an innocent person.)



a



b

FIGURE 4.35 Results of experiments to measure a sensory threshold using two different sets of instructions.

CONCEPT CHECK

14. Suppose we find that nearly all alcoholics and drug abusers have a particular pattern of brain waves. Can we now look for that pattern of brain waves and use it to identify people with an alcohol or drug problem? (Check your answer on page 152.)

Subliminal Perception

You have probably heard of **subliminal perception**, the idea that a *stimulus can influence our behavior even when it is presented so faintly or briefly or along with such strong distractors that we do not perceive it consciously*. (*Limen* is Latin for “threshold”; thus, subliminal means “below the threshold.”) Some people claim that subliminal perception can powerfully manipulate human behavior.

Are such claims plausible or outright nonsense? The first problem is to define *subliminal*. You already learned that *subliminal* means “below the *threshold*.” Fine, but you also learned that there is no sharp division between perceptible stimuli and imperceptible stimuli. If we define *subliminal* stimuli as those that people detect on fewer than 50% of occasions, then subliminal perception is hardly surprising. In practical terms, when psychologists refer to a “subliminal stimulus,” they generally mean “a stimulus that a person *did not* consciously detect on a given occasion,” regardless of whether the person could have detected it under other circumstances.

That definition does not solve the problem, however. How do we know whether someone *did* detect a given stimulus? We ask, of course. But a reply that “I did not see” the stimulus could mean “I saw nothing,” “I’m unsure what I saw,” or “I forget what I saw.” You can understand why this kind of research is often difficult to interpret.

What Subliminal Perception Cannot Do

Many of the claims regarding subliminal effects have been so outrageous and so lacking in evidence that many scientists came to doubt that subliminal perceptions had any effect at all. Many years ago, claims were made that subliminal messages could control people’s buying habits. For example, an unscrupulous theater owner might insert a single frame reading “EAT POPCORN” in the middle of a film. Customers who were not consciously aware of the message could not resist it, so they would flock to the concession stand to buy popcorn. Despite many tests of this claim, no one found any evidence to support it (Bornstein, 1989).

Another claim is that certain rock recordings contain “satanic” messages that have been recorded backward and superimposed on the songs. Some people allege that the listeners unconsciously perceive these messages and then turn to drugs or devil worship. As far as psychology is concerned, the issue is whether people who heard a backward message could understand it and whether it would influence their behavior. Psychologists have recorded various fairly tame messages (nothing satanic) and asked people to listen to them played backward. So far, no one listening to the backward messages has been able to discern what they would sound like played forward. And listening to those messages has not influenced anyone’s behavior in any detectable way (Vokey & Read, 1985). In other words, even if certain music does contain messages recorded backward, we have no reason to believe that these messages will influence the listeners.

A third unsupported claim: Many bookstores and music stores sell “subliminal audiotapes” that claim they can help you to improve your memory, quit smoking, lose weight, raise your self-esteem, and so forth. In one study, psychologists asked more than 200 volunteers to listen to a popular brand of audiotape. But they intentionally mislabeled some of the tapes. That is, some tapes with “self-